A PROJECT REPORT

ON

"SIGN LANGUAGE DETECTION USING HAND LANDMARK-BASED MACHINE LEARNING"

SUBMITTED TO

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA

IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF THE DEGREE

BACHELOR OF TECHNOLOGY

IN

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE

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CERTIFICATE

This is to certify that the project report titled **"Sign Language Detection Using Hand Landmark-Based Machine Learning"** is being submitted by **P. Afsana Khan (22KQ1A5423), G. Drakshavali (22KQ1A5411), S. Charan Kumar (22kq1a5457), G.Nithish Babu (22KQ1A5441), M. Vaishnavi (22KQ1A5420),** in **Bachelor of Technology in Department of Artificial Intelligence & Data Science**, is a record bonafide work carried out by them. The results embodied in this report have not been submitted to any other university for the award of any degree.

Project Guide **Head of the Department**

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DECLARATION

We hereby declare that the work is being presented in this dissertation entitled **"Sign Language Detection using Hand Landmark-Based Machine Learning"** is submitted towards the partial fulfillment of requirements for the award of the degree of Bachelor of Technology in Department of Artificial Intelligence & Data Science work carried out under the supervision of **Dr. G. T. Hari Krishna Murthy** M. Tech,(Ph.D),. PACE Institute of Technology and Sciences. The results embodied in this dissertation report have not been submitted by me for the award of any other degree. Furthermore, the technical details furnished in various chapters of this report are purely relevant to the above project and there is no deviation from the theoretical point of view for design, development and implementation.

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ABSTRACT

Communication between hearing-impaired individuals and the general public is often challenging because many people are unfamiliar with sign language. This project presents an assistive system that converts hand gestures into readable text using machine learning and computer vision techniques. The system captures hand gestures through a standard camera and processes them using real-time hand landmark detection. Instead of analyzing full images, the system extracts important hand key points and converts them into numerical features. These features are then provided to a trained machine learning classifier to recognize gestures representing common words or expressions. The predicted output is displayed as text on the screen instantly, enabling smooth and natural interaction. The proposed solution is simple, cost-effective, and user-friendly, as it requires no special hardware other than a camera. It can be effectively used in educational institutions, public service environments, and assistive communication applications. By translating visual gestures into text automatically, the system reduces dependence on human interpreters and promotes independent communication. The system is designed to operate under varying lighting conditions and different backgrounds, making it suitable for real-world scenarios. Furthermore, it can be extended in the future to support a larger vocabulary, continuous sentence formation, and integration with mobile or web-based platforms. Overall, this project demonstrates the practical potential of artificial intelligence in creating inclusive technologies and improving accessibility for the hearing-impaired community.

Key Terms:

Sign Language Recognition, Hand Gesture Recognition, Machine Learning, Computer Vision, MediaPipe, Random Forest, Assistive Technology.

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LIST OF ABBREVATIONS

Acronym	Abbreviations
AI	Artificial Intelligence
API	Application programming Interface
ML	Machine Learning
CV	Computer Vision
UI	User Interface
SVM	Support Vector Machine
KNN	K- Nearest Neighbors
RF	Random Forest
SLR	Sign Language Recognition
FPS	Frames Per Second
IDE	Integrated Development Environment
ROI	Region of Interest
RGB	Red Green Blue
NumPy	Numerical Python
OpenCV	OpenSource Computer Vision Library
MP	MediaPipe
CSV	Comma Separated Values
PKL	Pickle File Format

CHAPTER 1

INTRODUCTION

Communication is a fundamental part of human interaction, yet it remains a significant challenge for individuals who are hearing-impaired. Many people in the general public are not familiar with sign language, which creates barriers in daily conversations, education, healthcare, and public services. The lack of easily available interpreters further increases the communication gap, often limiting independence and social participation. With the rapid growth of artificial intelligence, there is an opportunity to design assistive technologies that can help bridge this divide and promote inclusive communication.

This project focuses on developing a real-time sign language detection system that converts hand gestures into readable text. The system uses computer vision techniques to capture hand movements through a standard camera and identify important hand landmarks. These landmarks are transformed into numerical representations, which are then processed by a machine learning classifier to determine the meaning of the gesture. The recognized output is displayed instantly on the screen, allowing users to communicate more effectively with people who do not understand sign language.

The proposed solution is designed to be simple, affordable, and practical. Since it requires only a camera and a computer, it can be deployed in classrooms, offices, and public environments without specialized equipment. By automating gesture recognition, the system reduces dependency on human interpreters and empowers users to express themselves more independently. Furthermore, the project demonstrates how artificial intelligence and vision-based techniques can be applied to create socially beneficial systems that enhance accessibility and digital inclusion.

System Components:

The Sign Language Detection system integrates multiple intelligent components to ensure accurate and real-time gesture interpretation. These components work together to capture hand movements, extract meaningful features, classify the gesture, and present the output in textual form.

Image Acquisition & Hand Landmark Detection – The system captures live video frames using a webcam and processes them through MediaPipe to identify the presence of a hand. The framework detects 21 key landmark points representing fingertips, joints,

and the wrist. These landmarks provide a compact and reliable representation of the hand structure, eliminating the influence of background noise and lighting variations. By focusing on geometric relationships rather than raw pixels, the system achieves faster and more stable recognition.

Feature Processing & Machine Learning Classification – Once landmarks are detected, their coordinates are normalized and converted into numerical feature vectors. These vectors are passed to a trained classifier implemented using Scikit-learn. The model analyzes spatial patterns in the hand structure and predicts the gesture class based on prior training. The predicted output is then mapped to a corresponding word or phrase using a label dictionary. Finally, the recognized text is displayed on the screen along with visual cues such as landmark drawings and bounding boxes, enabling smooth and user-friendly interaction.

Output and Integration

The system generates textual output representing the recognized hand gesture in real time. Extracted landmark features are evaluated by the classifier to determine the most probable interpretation. The result is displayed with visual cues, enabling quick and clear understanding. This structured output also supports integration with assistive communication platforms.

1.1. Background and Motivation

Communication is fundamental to human interaction, yet it remains a major challenge for individuals who are hearing-impaired when others are unfamiliar with sign language. In environments such as classrooms, workplaces, hospitals, and public service centers, the lack of interpreters often leads to misunderstandings and reduced independence. Many individuals must rely on writing or third-party assistance, which can be slow and inconvenient. This communication gap highlights the need for technological solutions that can translate gestures into a universally understandable form.

Recent advances in artificial intelligence and computer vision have enabled machines to recognize complex visual patterns with high reliability. Modern hand tracking frameworks can accurately detect key landmarks of the hand and transform them into numerical features. These features can then be analyzed using machine learning algorithms to identify specific gestures in real time. Such capabilities create new opportunities for building assistive systems that are fast, affordable, and practical for everyday use.

The motivation of this project is to design a real-time sign language detection system that converts hand gestures into readable text using only a standard camera. By leveraging landmark-based feature extraction and machine learning classification, the system aims to deliver accurate results while remaining simple and cost-effective. Ultimately, the project seeks to promote inclusive communication, reduce dependence on interpreters, and demonstrate the positive social impact that AI-driven assistive technologies can provide.

1.2. Problem Statement

Communication between hearing-impaired individuals and the general public remains a significant challenge because many people are not familiar with sign language. In everyday situations such as classrooms, workplaces, hospitals, and public service areas, the absence of interpreters often forces individuals to depend on writing or external help, which can be slow and inconvenient. This limitation reduces independence, causes delays in conveying important information, and creates barriers to smooth and natural interaction, highlighting the need for an automated system that can translate gestures into a universally understandable format.

Developing such a system, however, involves several practical difficulties. Variations in lighting, complex backgrounds, and differences in hand orientation can affect recognition accuracy, while some existing solutions require expensive hardware or heavy computation. Therefore, there is a need for a lightweight, cost-effective, and user-friendly approach that focuses on extracting essential hand features and applying efficient machine learning techniques to provide real-time predictions. A reliable system built with minimal equipment can help bridge the communication gap, improve accessibility, and promote inclusive participation for the hearing-impaired community.

In addition, such a solution can serve as a foundation for future assistive technologies aimed at improving human–computer interaction. It also opens opportunities for expanding gesture vocabularies and integrating intelligent communication support across multiple digital platforms.

1.3. Objectives of the Project

The primary objective of this project is to develop an intelligent sign language detection system that can translate hand gestures into readable text in real time. The system aims to use computer vision techniques to capture gestures through a standard camera and extract meaningful hand landmark features that represent the structure and movement

of the hand. By focusing on these essential points instead of full images, the project seeks to ensure faster processing, reduced computational complexity, and improved recognition accuracy. The solution is designed to operate with minimal hardware requirements so that it can be easily deployed in classrooms, workplaces, and public service environments.

Another important objective is to implement an efficient machine learning classifier capable of identifying predefined gestures based on the extracted features. The system should provide instant predictions and display them in a clear format that can be easily understood by individuals unfamiliar with sign language. Visual feedback, such as landmark drawings and bounding boxes, will also be incorporated to improve user interaction and confidence in the system's operation. Reliability under varying lighting conditions and different backgrounds is considered a key design goal to ensure practical usability.

Furthermore, the project aims to promote inclusive communication by reducing the dependency on human interpreters and enabling hearing-impaired individuals to interact more independently. The proposed framework is intended to be scalable, allowing the addition of more gestures, words, and sentence-level recognition in the future. Another objective is to make the system adaptable for integration with web and mobile applications, expanding accessibility across multiple digital platforms. Ultimately, the project demonstrates how artificial intelligence can be applied to create socially beneficial assistive technologies that bridge communication gaps within the community.

1.4 Scope and Applications

The scope of this project includes the design and implementation of an intelligent sign language detection system that translates hand gestures into readable text using computer vision and machine learning techniques.

The system is developed with an emphasis on real-time performance, usability, and minimal hardware requirements, ensuring that it can be accessed by a broad range of users. It captures live video input through a standard webcam and identifies important hand landmark points, which are then converted into numerical features for gesture recognition. By relying on landmark-based representation rather than full image processing, the system achieves faster computation and greater robustness against background variations, making it practical for everyday environments such as classrooms, offices, and public service centers.

To improve accuracy and reliability, the project applies a trained machine learning classifier that predicts predefined gestures based on patterns learned from collected data. The system is capable of providing instant feedback by displaying the recognized word or phrase directly on the screen, along with visual indicators such as hand skeleton drawings and bounding boxes. This immediate response helps users verify recognition results and enhances confidence in the interaction. The application is designed to function under different lighting conditions and camera positions, ensuring adaptability in real-world usage while remaining lightweight and easy to deploy.

The potential applications of this project are wide-ranging. It can serve as an assistive communication tool for hearing-impaired individuals, enabling smoother interaction with teachers, healthcare workers, customer service representatives, and the general public. The framework can also be integrated into educational platforms to support inclusive learning or embedded into kiosks and service systems where quick communication is essential.

1.5. Applications of the Proposed System

The proposed sign language detection system has a wide range of practical applications, particularly in situations where smooth and immediate communication is essential. It can act as a supportive communication bridge between hearing-impaired individuals and people unfamiliar with sign language, making it highly valuable in public places, educational institutions, offices, and healthcare environments. By converting gestures into clear textual output, the system helps users express their needs effectively and allows others to respond appropriately. This promotes greater independence and reduces misunderstandings in everyday interactions.

In institutional settings, the system can function as an assistive tool for teachers, service providers, and administrative staff by enabling faster and more inclusive communication. It can help during inquiries, learning activities, or emergency situations where rapid understanding is required. The application is also beneficial for elderly or non-technical users because it provides simple visual and textual feedback, making interaction intuitive and user-friendly. By offering real-time translation, the system can enhance participation and ensure that communication barriers do not limit opportunities.

Beyond direct communication support, the project serves as an AI-driven accessibility solution that promotes awareness and inclusivity. It can be used as a learning aid for

individuals who wish to understand sign language better by observing gesture-to-text mapping. Furthermore, the framework has strong potential for integration with mobile devices, web platforms, public kiosks, and customer service systems, expanding its reach and practical impact. As the vocabulary and recognition capabilities grow, the system can evolve into a comprehensive assistive communication platform for diverse communities.

1.6 Organization of the Report

This report is organized into multiple chapters to explain the project clearly, starting from the introduction and background to implementation and performance evaluation. The report is structured as follows:

Chapter 1 – Introduction

This chapter presents the project background, motivation, problem statement, objectives, scope, applications, and the overall structure of the report.

Chapter 2 – Literature Review

This chapter discusses existing studies related to sign language recognition, computer vision techniques, hand landmark detection, and machine learning–based gesture classification. It also identifies limitations in current systems and the research gap addressed by the proposed work.

Chapter 3 – Problem Formulation and Methodology

This chapter defines the problem and explains the proposed approach. It includes system architecture, module design, feature extraction using landmarks, workflow diagrams, and algorithms applied for gesture recognition.

Chapter 4 – Design and Implementation

This chapter describes the practical implementation using Python. It covers libraries used, dataset preparation, model training, and the logic behind real-time prediction and user interaction.

Chapter 5 – Results and Performance Analysis

This chapter presents the experimental outputs, accuracy evaluation, confusion matrix, and performance analysis based on speed, reliability, and usability.

Chapter 6 – Conclusion and Future Scope

This chapter summarizes the work, key outcomes, system limitations, and possible future enhancements such as expanding gesture vocabulary, dynamic recognition, and deployment on web or mobile platforms.

CHAPTER 2

LITERATURE REVIEW

Artificial Intelligence and computer vision have enabled the development of automatic sign language recognition systems to assist hearing-impaired individuals. Early research by Starner and Pentland [1] demonstrated the feasibility of vision-based gesture recognition using video input, forming the foundation for modern sign language detection systems.

Zhang et al. [2] showed that Convolutional Neural Networks (CNNs) can effectively recognize hand gestures from images. Their work proved that deep learning models automatically extract important features, improving recognition accuracy in image-based gesture systems.

However, CNN-based approaches require large labeled datasets and high computational resources. Molchanov et al. [3] highlighted that deep learning-based gesture recognition systems achieve high accuracy but demand powerful hardware, which limits their suitability for lightweight real-time applications.

To reduce computational complexity, MediaPipe Hands introduced by Zhang et al. [4] enables real-time detection of 21 hand landmarks using a lightweight model. Landmark-based representation improves efficiency by using coordinate points instead of full images, making it suitable for real-time gesture recognition.

Pugeault and Bowden [5] emphasized that feature extraction plays a crucial role in improving gesture recognition accuracy. Their research supports the use of structured hand features instead of raw image inputs for better performance.

Breiman [6] proposed the Random Forest algorithm, which provides high accuracy and robustness in classification tasks. It is particularly effective for structured numerical data such as hand landmark coordinates.

Pedregosa et al. [7] introduced the Scikit-learn library, which provides reliable tools for implementing machine learning models including Random Forest, Support Vector Machines, and Decision Trees for practical applications.

Bradski [8] introduced OpenCV, a widely used computer vision library for real-time image processing and video capture. It enables the practical implementation of live gesture detection systems.

Koller et al. [9] studied challenges in sign language recognition and emphasized the importance of efficient modeling techniques for real-time systems, especially in

handling dynamic gestures.

Recent surveys by Rastgoo et al. [10] highlight that lightweight, camera-based systems combining hand landmark extraction and traditional machine learning classifiers provide a balance between accuracy and computational efficiency.

Overall, the literature supports the use of real-time hand landmark detection combined with machine learning classification for developing efficient and cost-effective sign language recognition systems. The proposed project builds on these approaches to implement a practical real-time sign language detection system.

2.1 Review of AI Related Works

Artificial Intelligence (AI) has been widely applied in assistive communication systems to support automatic sign language recognition. Early approaches relied on image segmentation and rule-based classification, but they were sensitive to lighting conditions and background variations. The introduction of machine learning improved system flexibility by enabling gesture classification based on extracted features rather than predefined rules, thereby enhancing robustness and adaptability in different environments.

With advancements in deep learning, Convolutional Neural Networks (CNNs) significantly improved gesture recognition accuracy by automatically learning spatial features from images. However, CNN models require large annotated datasets and high computational power, which can limit their use in lightweight and real-time applications. To overcome this limitation, landmark-based methods such as MediaPipe Hands detect 21 three-dimensional hand landmarks and represent gestures as coordinate points, reducing computational complexity and minimizing background noise.

Recent research emphasizes integrating efficient machine learning classifiers such as Random Forest and Support Vector Machines with landmark-based feature extraction to achieve a balance between accuracy and performance. Combined with real-time computer vision libraries like OpenCV for video capture and preprocessing, these AI techniques enable the development of cost-effective, real-time sign language detection systems suitable for practical deployment and assistive communication applications.

Recent advancements in real-time computer vision have further improved the practicality of sign language recognition systems. Optimized frameworks and lightweight models allow gesture detection and classification to be performed directly through standard webcams without requiring specialized hardware. This makes AI-

based sign language systems more accessible and affordable for everyday use, especially in educational and assistive environments.

Moreover, current research focuses on improving system accuracy through better data preprocessing, normalization of hand landmarks, and model evaluation techniques. Performance metrics such as accuracy, precision, and recall are commonly used to validate model effectiveness. By combining real-time hand tracking, efficient feature extraction, and reliable machine learning classification, modern AI-driven systems provide a scalable and user-friendly solution for bridging communication gaps, forming the foundation of the proposed project.

2.2 Existing Techniques and Models

Sign language recognition systems use various techniques based on accuracy and real-time requirements. Early approaches relied on traditional image processing methods such as color segmentation and contour detection combined with rule-based classification. These systems were simple and computationally lightweight but were sensitive to lighting conditions, background variations, and changes in hand orientation.

With the advancement of machine learning, feature-based models such as Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), and Random Forest were introduced. These methods classify gestures using extracted features like hand shape descriptors or landmark coordinates. Random Forest, in particular, provides good accuracy and robustness for structured numerical data, though performance depends on effective feature extraction.

Deep learning techniques, especially Convolutional Neural Networks (CNNs), further improved gesture recognition by automatically learning spatial features from images. However, CNN models require large datasets and high computational resources. To improve efficiency, modern systems use landmark-based frameworks like MediaPipe Hands, which extract 21 key hand points. Combining landmark extraction with efficient classifiers enables accurate and real-time sign language detection, forming the basis of the proposed system.

These approaches help balance computational efficiency and recognition accuracy in real-time environments. The proposed project builds upon landmark-based detection and machine learning classification to develop a practical and cost-effective sign language recognition system.

2.3 Knowledge Base and Retrieval-Based Systems

Landmark-based systems represent a modern and efficient approach for vision-driven gesture recognition. Instead of analyzing complete images, these systems detect important key points of the hand such as fingertips, joints, and the wrist, and use their spatial relationships to identify gestures. Frameworks like MediaPipe provide highly optimized models that can extract these landmarks in real time using standard cameras. Because only coordinates are processed, computational requirements are significantly reduced, making such systems suitable for low-cost and portable deployments. These methods are reliable for known gestures, work effectively in varying backgrounds, and provide consistent numerical representations that can be easily used by machine learning algorithms. However, their performance depends on accurate landmark detection, and unusual hand poses or occlusions may affect results.

a. Feature Normalization and Vector Representation

After landmark detection, the raw coordinates must be transformed into a format that allows consistent classification. A common approach is normalization, where relative distances are calculated by subtracting minimum coordinate values. This ensures that the model focuses on the shape of the gesture rather than its position in the frame. The output becomes a structured feature vector that can be directly supplied to classifiers. The major advantage of this method is robustness against movement, camera shifts, and scale variations. However, it requires that preprocessing during prediction must match exactly with the preprocessing used during training.

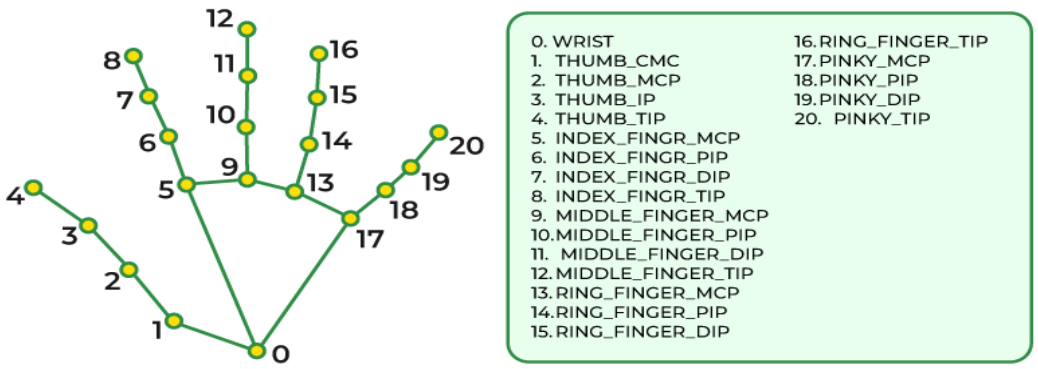

Fig 2.3 *Feature extraction from hand landmarks*

b. Machine Learning–Based Classification

Once features are generated, classification algorithms are used to map them to predefined gesture labels. Traditional supervised learning methods have shown strong performance in structured numeric datasets. Ensemble techniques, particularly those implemented in Scikit-learn, are popular due to their accuracy and efficiency. These models are trained on labeled landmark data and can produce predictions almost instantly, enabling real-time interaction. Compared to deep neural networks, such classifiers require less training data and computational resources, which makes them ideal for academic prototypes and deployable assistive tools.

The advantages of machine learning classifiers include fast execution, ease of training, and reliable performance on limited datasets. However, they may struggle when gestures outside the trained vocabulary are presented, which necessitates future dataset expansion.

c. Real-Time Interaction Systems

Modern literature also stresses the importance of usability in assistive applications. Real-time visualization, including bounding boxes and skeleton overlays, helps users align their gestures correctly and builds trust in the system. Immediate textual display allows natural communication between users and observers. Lightweight camera-based implementations are particularly valued because they eliminate the need for expensive hardware while maintaining interactive speeds. Such systems are increasingly used in inclusive education, service counters, and smart interfaces where accessibility is essential.

2.4. Limitations of Existing Approaches

Although numerous sign language recognition systems have been proposed in recent years, many still present limitations that reduce their effectiveness in real-world communication. A common issue is the reliance on specialized hardware such as sensor gloves or motion tracking devices, which increases cost and restricts accessibility for everyday users. Some vision-based approaches attempt to process entire images, making them computationally heavy and sensitive to background noise, lighting variations, and camera quality. As a result, these systems may fail to provide stable performance outside controlled laboratory environments.

Another major challenge is the limited adaptability of many existing models. Several implementations depend on large and complex deep learning architectures that require extensive training data and high-end computational resources. Such requirements make deployment difficult in educational institutions or public service settings where simple and affordable solutions are preferred. Additionally, many systems recognize only a small set of gestures and struggle when users present variations in hand orientation, speed, or style, reducing practical usability.

User interaction and feedback are also areas where earlier approaches often fall short. Without clear visual guidance such as landmark overlays or bounding boxes, users may find it difficult to understand whether the system is correctly detecting their gestures. Furthermore, delays in prediction or inconsistent outputs can reduce trust and make communication less natural. When gestures fall outside the trained vocabulary, most systems fail to provide meaningful alternatives, leaving the user without assistance.

Collectively, these limitations highlight the need for lightweight, real-time, and user-friendly solutions that can operate using standard cameras while maintaining reliable accuracy. Addressing these challenges is essential for creating assistive technologies that are practical, affordable, and capable of supporting inclusive communication in everyday environments.

2.5 Research Gap Identification

The literature survey reveals that many existing sign language recognition systems concentrate either on high-accuracy laboratory prototypes or on complex deep learning architectures that require significant computational resources. While these approaches achieve promising results, they often overlook the need for lightweight, affordable, and easily deployable solutions for everyday communication. Several systems depend on specialized sensors or expensive hardware, limiting accessibility for common users. Even camera-based methods sometimes process entire images, which increases computation time and reduces robustness in environments with background clutter or lighting variations.

Another important limitation identified in prior work is the lack of balance between performance and usability. Many models require very large datasets, powerful GPUs, or cloud infrastructure, which may not be feasible in educational institutions or public service applications. Furthermore, insufficient attention is often given to real-time feedback mechanisms that help users adjust their gestures. Without intuitive visual cues or immediate textual output, interaction becomes less natural and user confidence in the

system decreases.

Therefore, a clear research gap exists for systems that combine efficient hand landmark extraction, low computational cost, and reliable machine learning classification within a real-time framework. There is a strong need for solutions that can operate using standard cameras, provide instant and interpretable results, and remain practical for deployment in diverse real-world settings. The proposed project addresses this gap by integrating optimized landmark detection with structured feature processing and fast classification techniques, enabling accurate gesture-to-text translation without demanding expensive infrastructure.

By prioritizing simplicity, portability, and accessibility, the system aims to create an assistive communication tool that can be widely adopted. The framework also provides flexibility for future expansion, including support for additional gestures and more advanced interaction capabilities, while maintaining its lightweight design.

CHAPTER 3

PROBLEM FORMULATION & METHODOLOGY

3.1. Problem Definition

This chapter presents the problem formulation and methodology for the proposed project, titled Sign Language Detection using Hand Landmark–Based Machine Learning. The primary objective is to develop an intelligent assistive system capable of translating hand gestures into readable text in real time. Despite advancements in computer vision, many existing systems still depend on expensive hardware, complex deep learning architectures, or controlled environments, which limit their usability in everyday situations. Variations in lighting, background clutter, and differences in gesture presentation often reduce accuracy and make deployment challenging for practical communication scenarios.

To overcome these issues, the proposed system adopts a vision-based approach that relies on efficient hand landmark detection and lightweight machine learning classification. Instead of analyzing entire images, the system extracts key coordinate points representing the structure of the hand and converts them into normalized feature vectors. These features are then used to predict predefined gestures with minimal computational cost. The design ensures fast execution, making the solution suitable for real-time interaction using only a standard webcam.

The system is intended to provide immediate textual output along with visual feedback, enabling users to verify predictions and adjust their gestures if necessary. Unlike heavy deep learning methods that require large datasets and high-performance hardware, the adopted methodology emphasizes simplicity, portability, and ease of deployment. Overall, the problem addressed by this project is the need for an affordable, reliable, and user-friendly communication tool that can bridge the gap between hearing-impaired individuals and the wider community through accurate gesture-to-text translation.

3.2 Dataset Description

The Sign Language Detection system relies on a custom-built gesture dataset that serves as the foundation for training and evaluating the machine learning model. The dataset is designed to represent different hand signs corresponding to specific words or expressions used for communication. It is collected using a standard webcam, ensuring that the data reflects practical real-world conditions rather than controlled laboratory environments. This approach allows the system to generalize better during live prediction and makes it suitable for everyday deployment.

The dataset is organized in a structured directory format, where each folder corresponds to a particular gesture class. Images captured for each class are processed to extract hand landmarks, which are then converted into numerical feature vectors. Each sample is therefore represented not by raw image pixels but by the relative positions of key hand points. This structured numerical representation ensures consistency between training and prediction while significantly reducing computational requirements. Labels associated with each feature vector indicate the intended meaning of the gesture, allowing supervised learning algorithms to map patterns to their respective outputs.

The dataset includes multiple samples for every gesture to capture variations in orientation, distance from the camera, and minor differences in hand posture. Such diversity helps improve model robustness and prediction reliability. Additionally, the dataset is scalable, enabling the addition of new gestures or vocabulary in future updates. By focusing on landmark-based data rather than full images, the system achieves faster training, lower memory usage, and improved suitability for real-time applications. This dataset forms the backbone of the learning process and directly influences the accuracy and effectiveness of the final model.

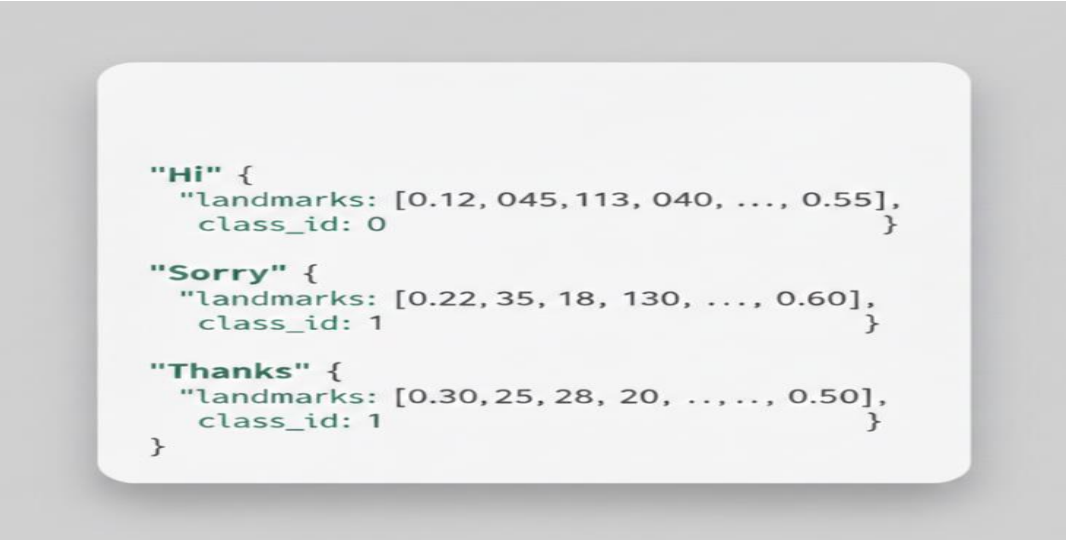

Fig 3.2 *Dictionary style representation of gesture features and labels used for training*

3.3 Data Preprocessing Technique

The data preprocessing module plays a critical role in preparing captured hand gesture inputs for accurate recognition. Since raw video frames may contain background noise, lighting variations, and irrelevant visual details, preprocessing ensures that only meaningful hand information is supplied to the classifier. By transforming images into structured numerical representations, the system improves consistency between training and prediction while reducing computational complexity. Effective preprocessing is therefore essential for reliable real-time gesture interpretation.

a. Image and Landmark Preprocessing

Image preprocessing begins by capturing frames from the webcam and converting them into a suitable color format for hand tracking. The system then detects the presence of a hand and extracts key landmark points representing fingertips, joints, and wrist positions. These coordinates are collected and organized into arrays that describe the spatial structure of the gesture. Instead of using raw pixel values, this approach focuses only on geometric information, which makes the system more robust to background variations.

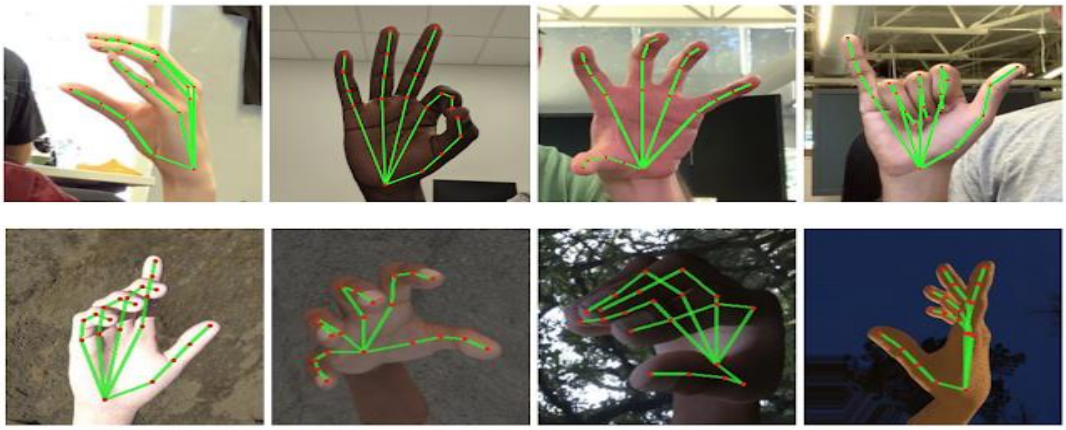

Fig3.3.1 *Preprocessing stage involving frame capture, color conversion, and landmark detection*

b.Image Input Preprocessing

For vision-based gesture recognition, the preprocessing module prepares live video frames captured from the webcam to meet the requirements of the hand tracking system. The application continuously reads frames and verifies successful capture before further processing. Since different libraries operate on specific color spaces, the system converts the default BGR image format into RGB to ensure compatibility with the hand detection framework. This step is essential because improper color representation can significantly reduce detection accuracy.

Once converted, each frame is analyzed to determine whether a hand is present. If landmarks are detected, the coordinates are extracted for subsequent feature generation; otherwise, the system safely skips the frame to prevent invalid predictions. Optional visual enhancements, such as drawing landmark connections, may be applied to help users confirm that the hand is correctly identified. By validating input quality and standardizing frames before classification, this preprocessing stage ensures smooth operation and stable real-time performance.

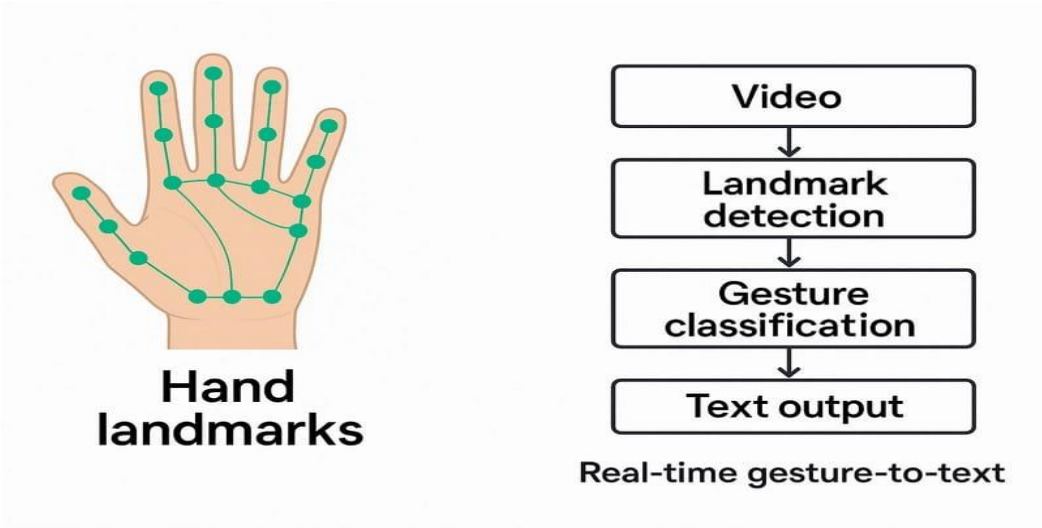

Fig 3.3.2 *Webcam frame preprocessing and hand landmark detection prior to feature extraction*

c. Error Handling and Logging

To enhance system reliability, the preprocessing module integrates error handling and monitoring mechanisms throughout real-time operation. The application continuously verifies that frames are successfully captured from the webcam and checks whether valid hand landmarks are detected before forwarding data for prediction. If a frame is unreadable or no hand is present, the system skips the input and continues processing subsequent frames, preventing incorrect outputs or interruptions. Detection events and failures can be observed through on-screen visual feedback, which assists in troubleshooting and improving user interaction.

d. Integration with Downstream Modules

After preprocessing, the validated hand landmark data is forwarded to the feature extraction and classification stages for gesture interpretation. By ensuring that only reliable frames are processed, the module prevents errors from propagating through the system. The standardized coordinate format matches the structure used during model training, which is essential for accurate prediction. Early removal of noisy or invalid inputs reduces computational overhead and improves response time

Outcome:

After preprocessing, each valid frame is converted into structured landmark data ready for feature generation and classification. Frames without reliable detection are excluded to maintain prediction accuracy. This stage forms the foundation for stable and real-time gesture-to-text conversion.

3.4. Feature Extraction and Selection

The feature extraction module is a crucial component of the sign language detection system, acting as the bridge between detected landmarks and gesture prediction. Its main function is to convert raw coordinate points into structured numerical representations that can be interpreted by the classifier. Instead of processing full images, the system focuses on geometric relationships among hand key points. This approach reduces noise, improves efficiency, and ensures consistent input for the model. By standardizing how gestures are represented, the module enables accurate and reliable real-time recognition.

a.Text Feature Extraction

In the proposed system, gesture data replaces traditional textual inputs and becomes the primary source of information for interpretation. Instead of words or sentences, the application relies on hand landmark coordinates obtained from each captured frame. These points correspond to important regions such as fingertips, joints, and the wrist, and together they describe the configuration of the hand during a gesture. This structured representation allows the system to treat physical movements as machine-readable input.

Following detection, the coordinates are collected and arranged systematically to preserve spatial relationships among different parts of the hand. Each landmark contributes to the overall gesture identity, and even small positional variations can influence classification. By representing gestures numerically, the module enables efficient comparison with previously learned patterns and ensures suitability for automated recognition.

This transformation from visual input to structured numerical data provides the basis for accurate classification. It allows the model to analyze gestures consistently across different users and environments while maintaining real-time performance. As a result, the feature extraction process becomes a critical step in converting human motion into meaningful textual output.

b.Image Feature Extraction

In the proposed system, visual input from the webcam serves as the primary source for gesture understanding. Each captured frame is first verified for successful acquisition and then analyzed to determine whether a hand is present. Upon detection, the system identifies key landmark points corresponding to fingertips, joints, and the wrist. These landmarks provide a concise yet informative description of the gesture while avoiding

unnecessary background details.

The extracted points are represented through their x and y coordinates, which together capture the spatial configuration of the hand. This numerical representation enables the system to distinguish subtle differences between gestures that may appear similar to the human eye. Because only landmark data is processed, the computational burden is significantly reduced, supporting efficient real-time performance.

By converting visual information into structured coordinate data, the module produces features that can be directly interpreted by the machine learning classifier. This transformation ensures consistency between training and prediction phases while enhancing robustness across variations in lighting, distance, and user style. As a result, the system can reliably translate hand movements into meaningful textual output.

c.Additional Enhancements and Reliability Measures

To maintain robust performance, the feature extraction module integrates validation and consistency checks throughout the detection process. Landmark coordinates are considered only when a hand is confidently identified; otherwise, the frame is ignored to prevent incorrect predictions. The system ensures that the number and order of extracted points remain consistent with the structure used during training, guaranteeing compatibility with the classifier. Visual overlays of landmarks provide immediate confirmation to users and assist in correcting hand placement when detection is uncertain.

The module is also designed with reliability in mind by gracefully handling ambiguous or incomplete inputs. If landmarks are partially detected or tracking confidence is low, prediction is temporarily avoided rather than risking inaccurate output. These safeguards reduce false classifications, improve stability, and enhance user trust in the system.

Outcome and Significance

By the end of the feature extraction stage, the system produces a structured numerical description of the detected hand gesture. The raw visual input is converted into organized landmark coordinates that accurately represent the geometry of the hand. These standardized features are then ready to be processed by the trained classifier for prediction.

Ensuring uniform formatting between training and real-time data greatly improves accuracy and stability. The transformation reduces background influence and focuses only on meaningful gesture information. This makes the system efficient, lightweight, and suitable for continuous operation. Ultimately, the module enables reliable gesture-to-text conversion and strengthens the overall effectiveness of the assistive communication system.

3.5. Proposed Methodology

The proposed sign language detection system is designed to address the limitations of traditional gesture recognition approaches that often depend on specialized hardware, computationally intensive models, or controlled environments. The project adopts a lightweight vision-based strategy by combining real-time hand landmark detection with efficient machine learning classification. Video frames captured from a standard webcam are processed to extract key coordinate points, which are normalized and converted into structured features for prediction. The trained model then determines the most probable gesture and maps it to a readable word or phrase, displayed along with visual feedback for user confirmation. By emphasizing speed, simplicity, and affordability, the methodology enables practical deployment in everyday environments while providing a scalable foundation for future expansion.

Key Features of the Proposed System:

a.Vision-Based Gesture Analysis

The proposed system interprets sign language gestures using real-time visual input captured from a standard webcam. Each frame is processed to detect the hand and identify key landmarks such as fingertips, joints, and wrist positions. These points are transformed into structured numerical features that represent the geometry of the gesture while minimizing the influence of background elements. The feature set is then evaluated by a trained machine learning classifier to determine the most probable meaning, and the result is displayed instantly as readable text along with visual feedback. This approach enables natural, fast, and equipment-free interaction, making the system practical for everyday assistive communication.

b. Machine Learning Model Integration

The trained machine learning model is integrated into the system to provide accurate and fast gesture interpretation. After hand landmarks are extracted and converted into

structured numerical features, the classifier analyzes the spatial patterns learned during training to predict the most probable gesture. This integration allows the system to deliver instant results while maintaining consistency between training and real-time usage. By relying on a lightweight yet powerful model, the approach balances accuracy, efficiency, and ease of deployment, overcoming the limitations of manual interpretation or hardware-dependent solutions.

c. Dataset - Driven Gesture Prediction

The custom gesture dataset forms the core of the system's learning capability. It consists of labeled samples representing different hand signs, where each instance is stored as a structured set of landmark coordinates. This dataset-driven strategy enables the model to learn spatial patterns associated with specific gestures and ensures fast, consistent predictions during real-time operation. Because the knowledge is derived from locally trained data, the system can function efficiently without dependence on external services. The approach also allows straightforward expansion by adding new gestures and retraining the model when required.

d. Robust Handling of Gesture Variations

To manage real-world variations in hand movement, the system is designed to tolerate small differences in finger positions, orientation, and distance from the camera. Instead of expecting perfectly identical gestures, the trained classifier recognizes patterns within a learned range of variation. This flexibility allows the model to correctly interpret signs even when users perform them with minor inconsistencies. Such robustness improves usability, reduces recognition failures, and enhances overall interaction quality, making the application practical for diverse users and environments.

e. Structured Output Generation

The system presents prediction results in a clear and consistent format by mapping the recognized gesture to a predefined word or phrase. The output is displayed directly on the screen along with visual cues for easy understanding. This structured presentation ensures uniform interpretation across executions and improves user confidence.

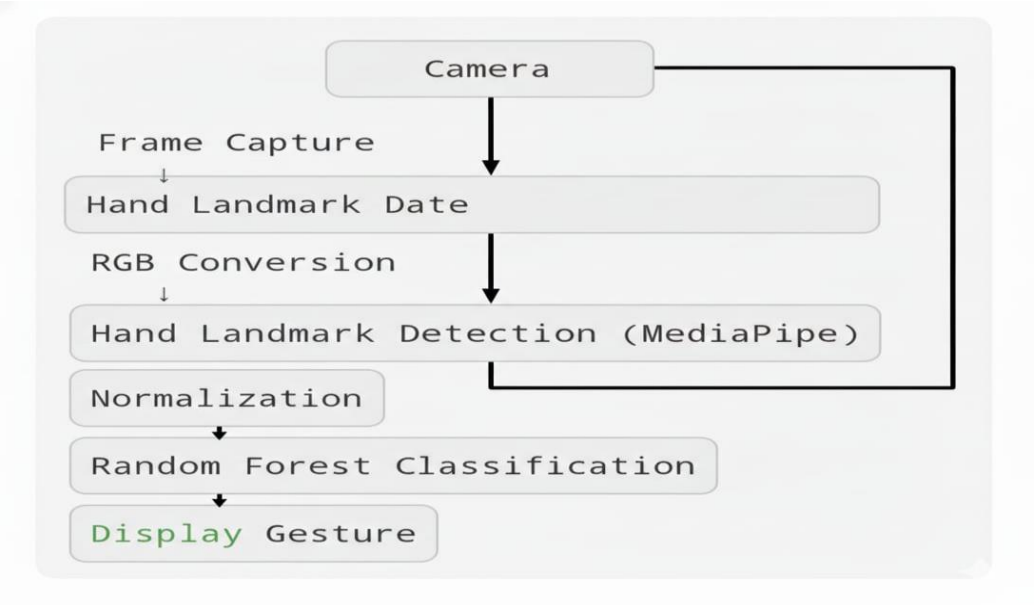

Fig 3.4 *Flowchart of real-time gesture recognition from video capture to text output.*

f. Image Preprocessing Using OpenCV

Video frames captured from the webcam are preprocessed using OpenCV to prepare them for accurate hand landmark detection. The system converts each frame from BGR to RGB format, which is required by the hand tracking model. Basic adjustments such as resizing and frame consistency help maintain stable detection performance. This preprocessing improves reliability and ensures that the model receives standardized input for further analysis. Errors in frame capture are handled gracefully, allowing continuous real-time operation.

g. User-Centric Communication Support

The system provides user-friendly communication assistance by translating hand gestures into clear textual output in real time. It helps bridge the interaction gap between hearing-impaired individuals and people who may not understand sign language. The interface is designed to be simple and transparent, allowing users to easily view predictions and adjust gestures when needed. Immediate feedback improves confidence and usability, making the application suitable for everyday conversations, learning environments, and assistive communication scenarios.

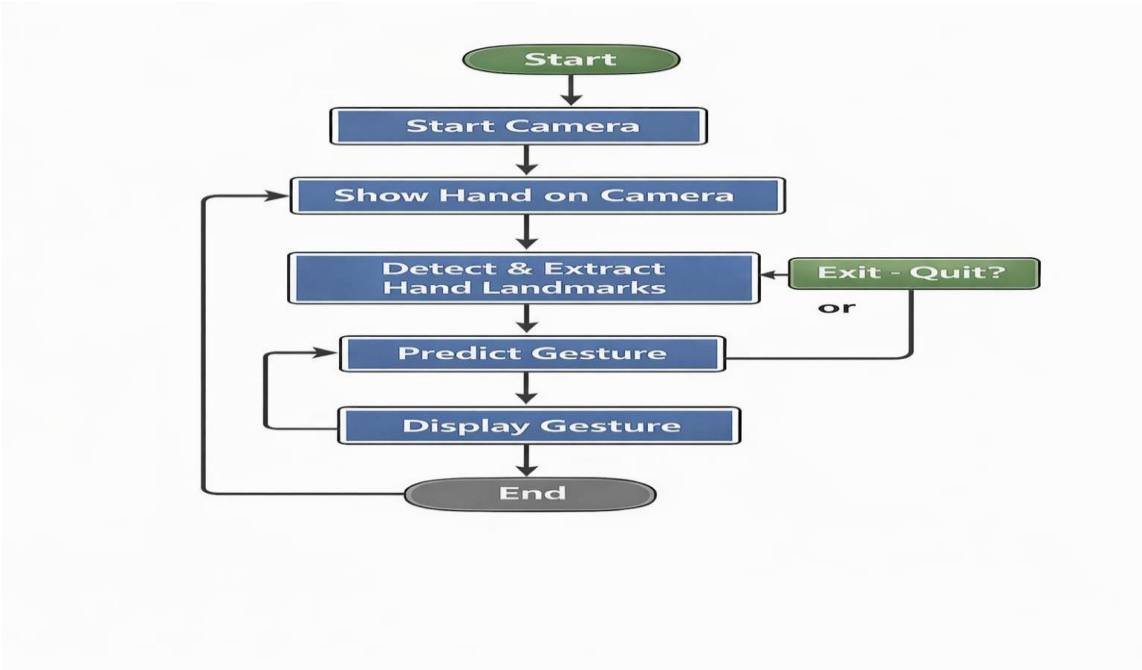

Fig 3.5 User Interaction Loop for Real Time Gesture Recognition

h. Scalability and Extensibility

The system is designed for scalability and future extensibility. It can be expanded to include additional diseases, improved clinical datasets, multilingual support, mobile app integration, and more advanced AI models. The hybrid approach ensures a balance of speed, accuracy, and intelligence while maintaining usability across diverse user scenarios. The system is designed with scalability and future extensibility in mind. It can be expanded to support additional gestures, larger training datasets, and improved classification techniques to enhance recognition accuracy. The framework also allows integration with mobile or web-based platforms, enabling wider accessibility for users.

i. Advantages and Limitations

The proposed system provides fast and reliable real-time gesture recognition using efficient hand landmark detection and machine learning classification. It works with a standard webcam, avoids the need for specialized hardware, and delivers immediate textual output that improves communication accessibility. The lightweight design enables smooth performance and makes the solution practical for everyday interactions.

3.6.System Architecture / Block Diagram

The system architecture consists of the following modules:

User Interface (Webcam Input)

Frame Capture Module

Image Preprocessing (RGB Conversion)

Hand Landmark Detection Module

Feature Normalization Module

Machine Learning Classification Module

Output Display Module

This architecture ensures speed, accuracy, and real-time responsiveness.

Architecture Flow:

Gesture Input Flow:

User → Webcam Interface → Frame Capture → RGB Conversion → Hand Landmark Detection → Feature Normalization → Random Forest Classifier → Display Predicted Gesture → User

Image-Based Gesture Flow

User → Webcam → Capture Frame → Convert BGR to RGB → Detect Hand Landmarks → Create Feature Vector → Predict Gesture → Display Text → User

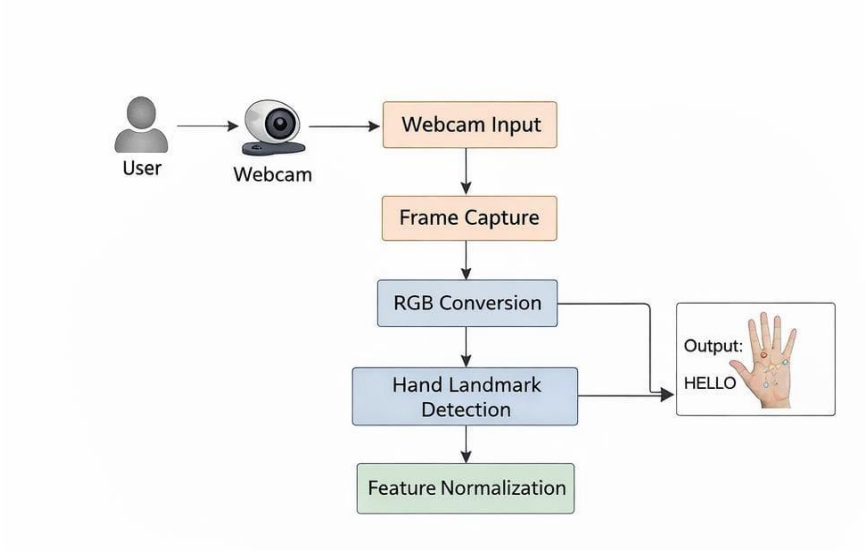

Fig 3.6 *Overall System Architecture of the Hand Gesture Recognition System*

3.6.1.UML DIAGRAM

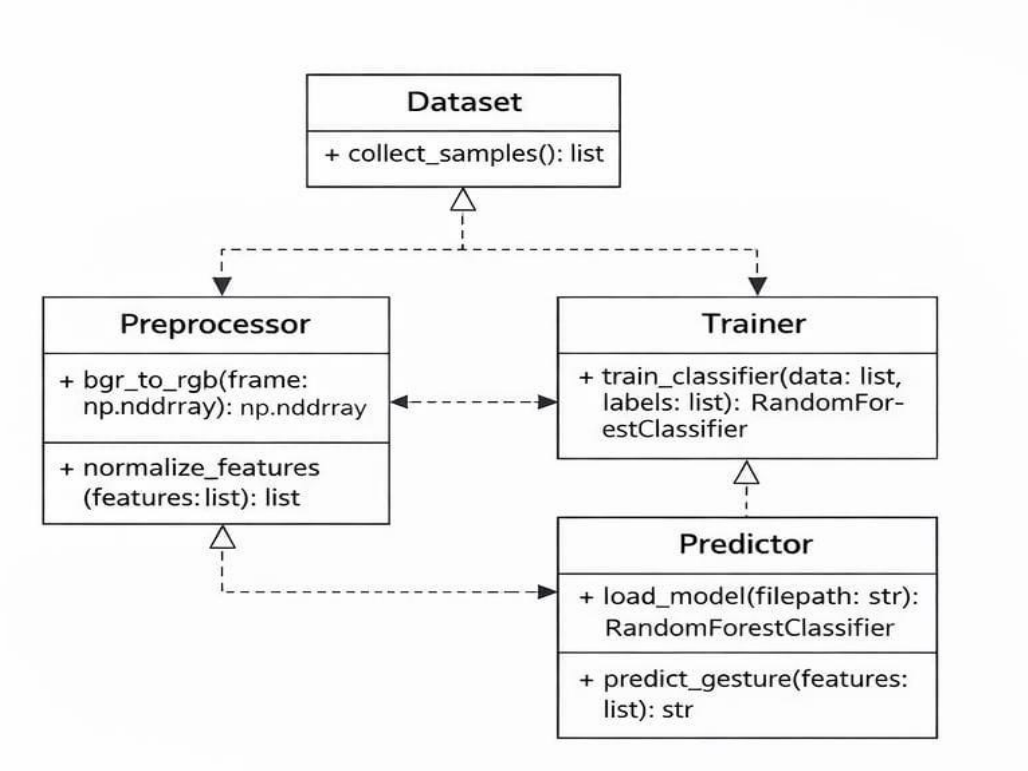

Fig 3.6.1 *UML Diagram of the Gesture Recognition Modules*

3.6.2.USE CASE

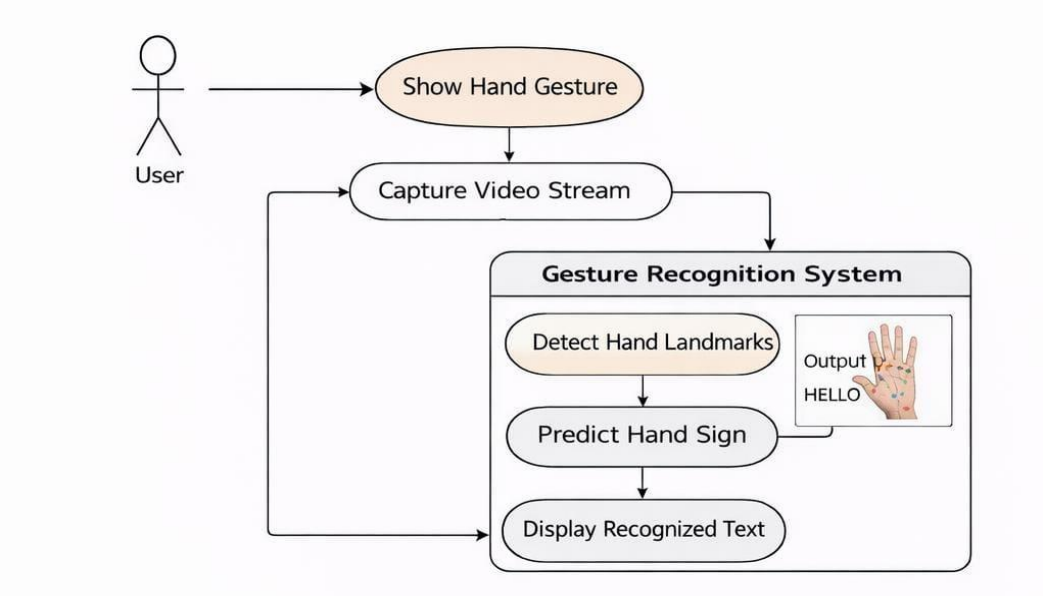

Fig 3.6.2 *Use Case Diagram of the Gesture Recognition System*

3.6.3.CLASS DIAGRAM

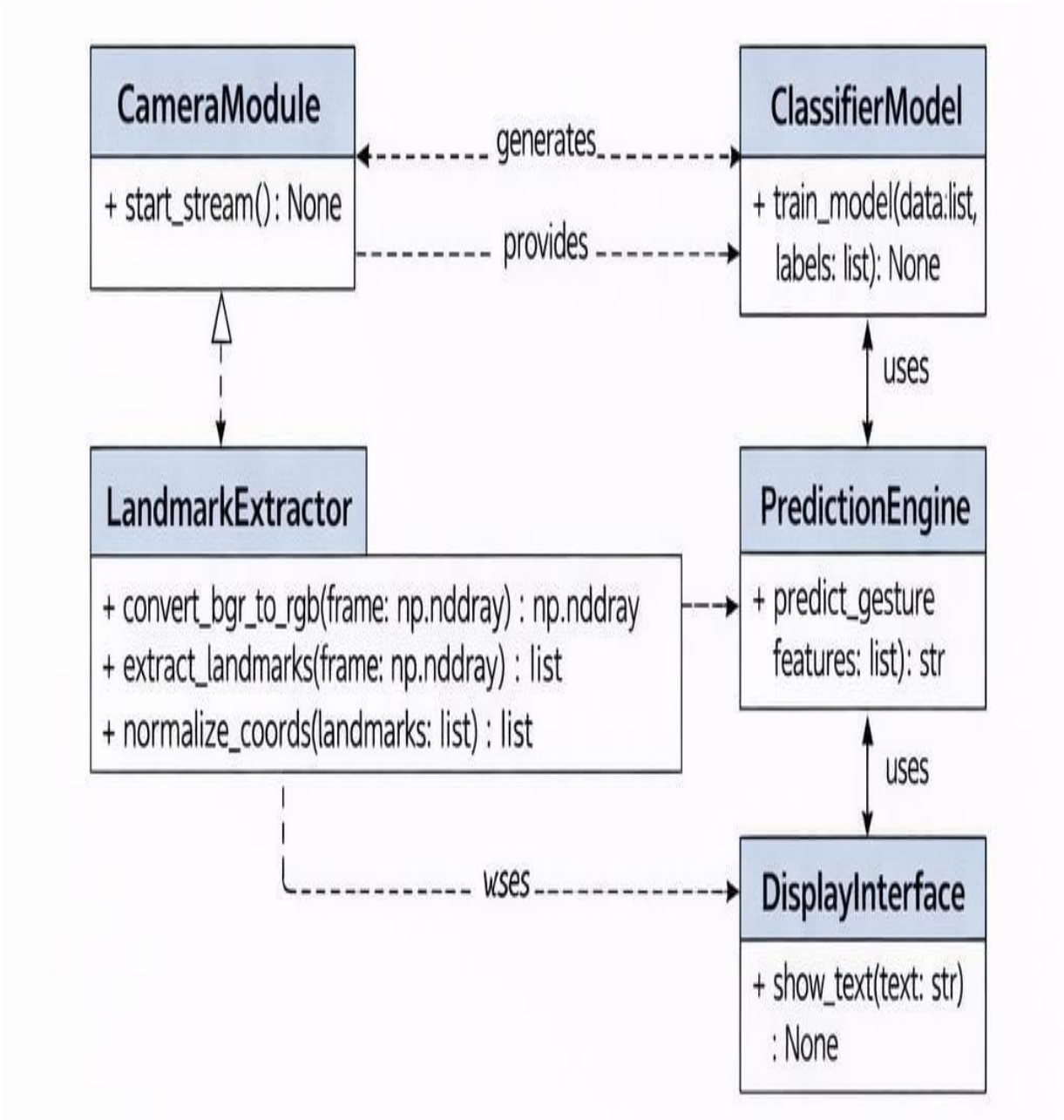

Fig 3.6.3 *Class Diagram of the Gesture Recognition System*

3.6.4 SEQUENCE DIAGRAM

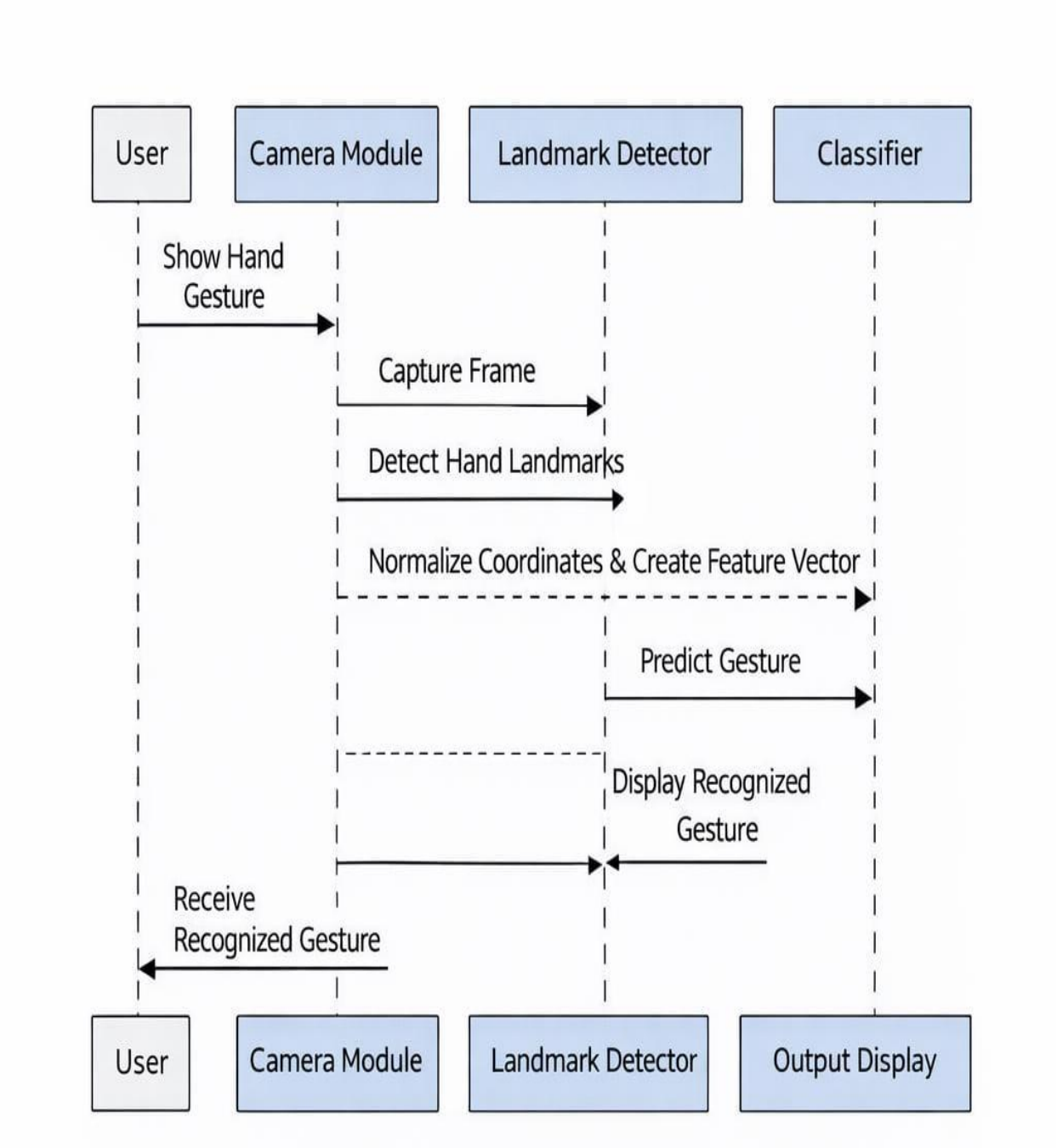

Fig 3.6.4 *Sequence Diagram of the Hand Gesture Recognition System*

3.6.5 ACTIVITY DIAGRAM

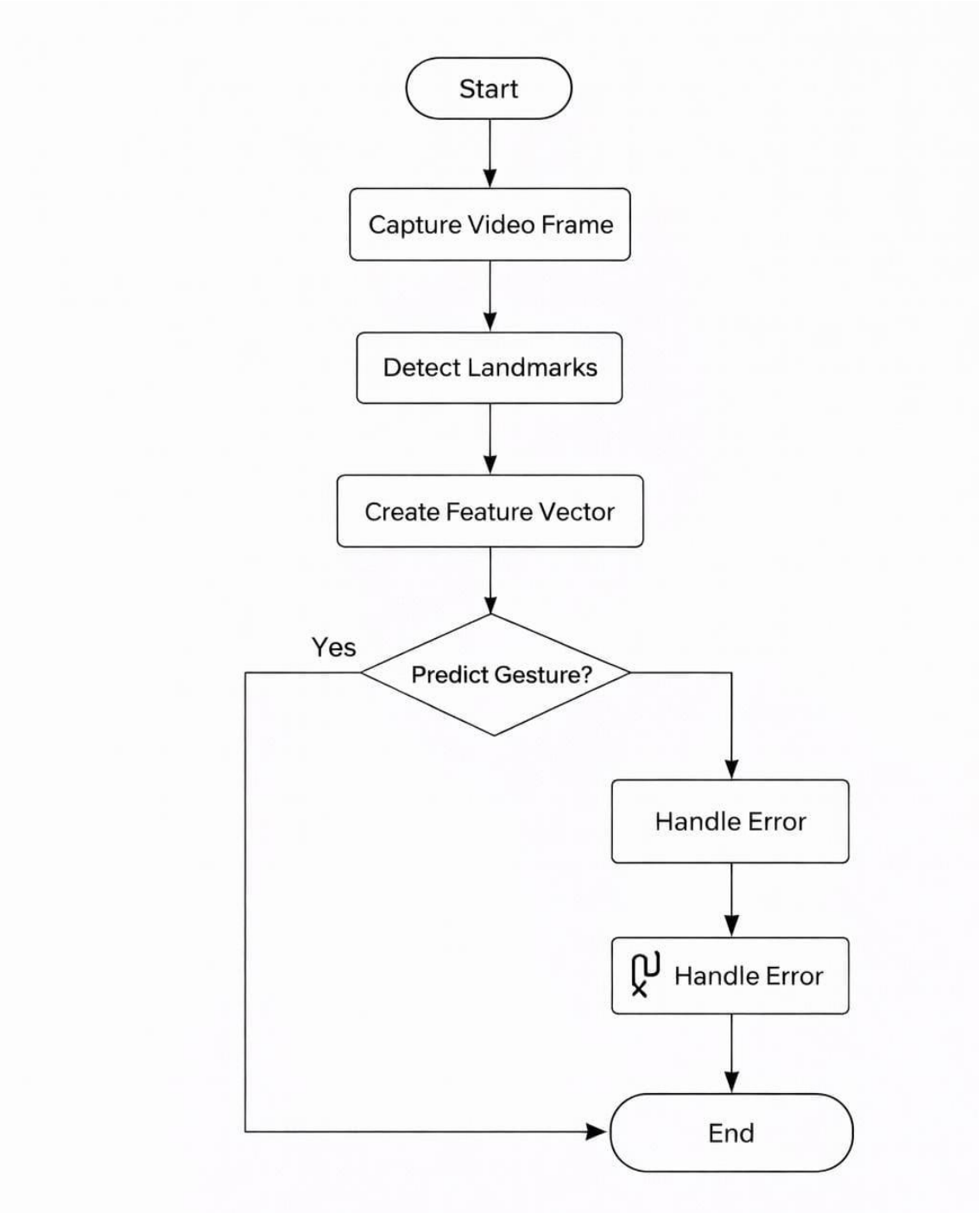

Fig 3.6.5 *Activity Diagram of the Gesture Recognition System*

3.7Algorithms / Flowcharts

Algorithm 1: **Real-Time Gesture Recognition (Webcam Input)**

1. Start
2. Initialize camera and load trained classification model
3. Capture video frame from webcam
4. Detect hand and extract landmarks using MediaPipe
5. Preprocess and normalize landmark coordinates
6. Send features to the classifier
7. Predict gesture label
8. Display predicted gesture on screen
9. Repeat until user exits
10. Stop

Algorithm 2:Image-Based Gesture Prediction

1. Start
2. Load input hand image
3. Detect hand region using MediaPipe
4. Extract landmark coordinates
5. Normalize and convert into feature vector
6. Pass features to trained model
7. Generate gesture class label
8. Display prediction
9. Stop

CHAPTER 4

MODEL DESIGN & IMPLEMENTATION

4.1.AI Models Used

This project involves the use of a traditional Machine Learning (ML) model for gesture classification rather than training large Deep Learning (DL) networks. The system combines real-time hand landmark detection with an efficient classifier that interprets spatial patterns of finger and joint positions. The landmark extraction framework provides reliable geometric representations of the hand, while the trained model performs prediction based on features derived from these coordinates. The main objective of using this AI approach is to deliver quick, structured, and user-friendly gesture recognition that works effectively in real-world environments. Since the model operates on compact numerical data instead of full images, the overall solution remains lightweight, scalable, and easy to deploy. This integration allows the system to handle continuous video input and produce immediate textual output without demanding high computational resources.

a.Machine Learning Classifier (Gesture Recognition Model)

The machine learning classifier acts as the core component of the sign language detection system, converting extracted landmark coordinates into meaningful gesture labels. By operating on structured numerical data rather than raw images, the approach reduces computational cost while maintaining reliable recognition. The model focuses on spatial relationships between fingers and joints, making it efficient for real-time applications. This design allows the system to remain lightweight and responsive even on standard hardware.

Trained on labeled examples, the classifier learns patterns associated with each gesture and compares them with incoming data during live operation. This results in fast, consistent, and interpretable predictions. The immediate display of results helps users verify recognition and adjust their gestures if needed. Overall, the method enables practical and accurate gesture-to-text translation without requiring specialized hardware.

b.Hand Landmark Detection Model

This hand landmark detection framework is integrated into the system to enable accurate interpretation of visual gestures from live video input. Many assistive communication scenarios require precise understanding of finger positions and hand structure, which cannot be achieved through simple image processing alone. The model identifies important key points such as fingertips, joints, and the wrist, allowing the system to analyze the geometric configuration of the hand. By extracting structured coordinates instead of raw pixel values, the method ensures efficient processing and reliable performance even in the presence of background variations.

This model plays a crucial role in transforming visual information into machine-readable data that can be used for classification. It continuously tracks the hand in real time and provides consistent landmark outputs that form the basis of gesture recognition. The structured representation improves robustness and enables the classifier to distinguish between similar gestures with higher accuracy. By integrating automated landmark detection, the system becomes more practical for everyday usage and accessible to users without requiring specialized equipment.

c. Hybrid Working Approach

The proposed system follows a hybrid strategy by combining real-time hand landmark detection with machine learning–based classification. The landmark module continuously tracks fingers and joints from live video input and converts them into structured coordinate data. By focusing only on geometric information rather than full images, the system achieves faster processing and reduced sensitivity to background variations. This makes the approach efficient and suitable for real-time assistive communication.

Once landmarks are extracted, the classifier evaluates these features to determine the most probable gesture. If detection is unstable, the system waits for clearer input instead of generating unreliable predictions, thereby improving accuracy and user trust. This collaboration between detection and classification balances speed, reliability, and practicality. The design also supports future scalability, allowing new gestures to be added through dataset expansion and retraining.

4.2.Training and Testing Process

The proposed Sign Language Detection system involves a supervised machine learning approach in which a classifier is trained using hand landmark features extracted through MediaPipe. During the training phase, gesture images were collected for each class using a webcam. For every captured frame, MediaPipe detected 21 hand landmarks, and their coordinates were converted into numerical feature vectors. These vectors, along with their corresponding gesture labels, formed the dataset used to train the classification model. This approach eliminates the need for raw image training and instead focuses on geometric hand representations, making the system lightweight and fast.

To build a reliable model, the dataset was divided into training and testing subsets. The training data was used to teach the classifier the relationship between landmark patterns and gesture labels, while the testing data was reserved for performance evaluation. The Random Forest algorithm was chosen because of its robustness, ability to handle non-linear data, and good performance with small-to-medium datasets. After training, the model was saved and later loaded during real-time prediction.

Testing was conducted using both pre-collected test samples and live webcam input. In real-time testing, users performed gestures in front of the camera, and the system detected the hand, extracted landmarks, normalized the coordinates, and passed them to the trained model. The predicted class label was then displayed on the screen. These experiments verified that the model could correctly recognize gestures under different lighting conditions, backgrounds, and hand orientations.

Special attention was given to evaluating how the system behaves in practical scenarios. Tests included partial hand visibility, fast movement, and variations between different users. The results showed that MediaPipe consistently tracked landmarks accurately, and the classifier produced stable predictions for most gestures. Minor errors occurred mainly when the hand was not clearly visible or when gestures were very similar.

To ensure reliability, the system was also tested for failure conditions, such as absence of a hand in the frame or detection confidence below threshold. In such cases, the system avoided false predictions and waited until a valid hand was detected. Overall, the training and testing process confirmed that the model is capable of providing accurate, real-time gesture recognition with low computational requirements, making it suitable for practical sign language assistance applications.

In addition to real-time evaluation, quantitative validation was performed using standard performance metrics. The predictions generated on the test dataset were compared with the actual gesture labels to compute accuracy. This helped in understanding how well the classifier generalizes to unseen data. The overall accuracy obtained during experiments indicated that landmark-based learning is effective for representing hand gestures without requiring computationally expensive image processing.

Another important part of testing focused on repeatability and consistency. The same gestures were performed multiple times by different users to check whether the system produced stable outputs. Despite variations in hand size, speed, and minor positional differences, the model was able to maintain consistent recognition in most cases. This demonstrated that the normalization and feature extraction techniques helped reduce user-specific bias.

Finally, system responsiveness was evaluated to ensure suitability for live interaction. The time taken for hand detection, landmark extraction, and prediction was measured across multiple trials. Results showed that the pipeline operates within real-time limits, with minimal delay between gesture performance and output display. This confirms that the proposed approach is efficient enough for practical deployment in sign language communication and human-computer interaction scenarios.

The evaluation also highlighted the importance of providing clear gestures within the camera frame for optimal recognition. Users quickly adapted to the system after brief interaction, indicating good learnability. The model demonstrated strong potential for assisting basic communication between hearing-impaired individuals and the general public. With additional training data, the recognition capability can be further improved. These results validate the feasibility of using landmark-based machine learning for real-time sign interpretation.

4.3 Hyperparameter Tuning

Unlike rule-based systems, the proposed Sign Language Detection project involves training a machine learning classifier; therefore, hyperparameter tuning plays an important role in improving prediction accuracy and stability. Although the system uses MediaPipe for landmark detection instead of raw image training, the performance of the final model depends heavily on how well the classifier parameters and detection thresholds are selected. Proper tuning ensures better generalization, reduced overfitting, and reliable real-time gesture recognition.

One of the key components tuned during development was the configuration of the Random Forest model. Parameters such as the number of decision trees, maximum tree depth, and splitting criteria were adjusted experimentally. Increasing the number of trees generally improved accuracy but also increased computation time, while deeper trees sometimes caused overfitting. Through repeated trials, balanced values were selected to achieve high recognition performance while maintaining real-time speed.

Another important aspect of tuning involved the confidence thresholds used by MediaPipe for hand detection and tracking. If the detection confidence was set too high, the system occasionally failed to recognize hands under challenging lighting or partial visibility. If it was too low, false detections could occur. By carefully selecting optimal confidence levels, the system achieved stable landmark extraction across different users and environments.

Feature normalization parameters were also refined to ensure that variations in hand size, distance from the camera, and orientation did not negatively affect predictions. By scaling landmark coordinates relative to reference points, the classifier became more robust and consistent. This step significantly improved recognition accuracy when different individuals performed the same gesture.

Finally, runtime behavior such as prediction smoothing and frame-by-frame updates was adjusted to avoid flickering outputs. Instead of reacting to single noisy predictions, the system stabilizes results over consecutive frames, providing a smoother user experience. Overall, hyperparameter tuning helped the model achieve reliable, fast, and accurate gesture recognition suitable for practical sign language communication.

4.4.Tools, Frameworks, and Libraries Used

The Sign Language Detection system is implemented using Python and designed to run on standard desktop or laptop environments with a webcam. Python was selected because of its simplicity, rich ecosystem of computer vision and machine learning libraries, and strong community support. The tools used in this project enable real-time video capture, hand landmark extraction, feature processing, model training, and live prediction. The overall setup ensures that the system remains lightweight, easy to deploy, and suitable for academic as well as practical applications.

Programming Language

Python 3.10+

Python is used due to its readable syntax, rapid development capability, and extensive support for artificial intelligence and computer vision. It allows seamless integration between data collection, preprocessing, model training, and real-time inference. Python also provides reliable libraries for numerical computation and visualization.

Development Environment

The project can be executed on any system supporting Python with a connected webcam. Development and testing were performed using standard IDEs and local runtime environments, ensuring flexibility without requiring specialized hardware.

Libraries Used

OpenCV (cv2)

OpenCV is used for capturing live video from the webcam, reading image frames, color space conversion, and displaying prediction outputs. It provides the foundation for real-time interaction between the user and the system.

MediaPipe

MediaPipe is responsible for detecting hands and extracting 21 key landmark points. These landmarks represent the geometric structure of the hand and are later converted into feature vectors for classification.

NumPy

NumPy is used for numerical operations, array manipulation, and preparing structured input for the classifier. It enables efficient mathematical computation required during normalization and prediction.

scikit-learn

The scikit-learn library provides the implementation of the Random Forest classifier used for gesture recognition. It supports model training, prediction, dataset splitting, and performance evaluation.

pickle

Pickle is used to save and load trained models. Once training is completed, the classifier is stored and reused during real-time execution without retraining.

matplotlib

Matplotlib is utilized for visualizing performance metrics such as accuracy and error trends during experimentation and evaluation.

4.5. Hardware and Software Requirement

The Sign Language Detection system requires a standard computing setup capable of handling real-time video capture, hand landmark extraction, and machine learning–based prediction. Since the application processes continuous webcam input and performs frame-by-frame analysis, the hardware must support moderate computational workloads. The following specifications are recommended to ensure smooth execution during both development and deployment.

Processor (CPU)

A multi-core processor is required for video processing, landmark computation, and classification tasks. The minimum requirement is an Intel Core i3 or AMD Ryzen 3, while an Intel Core i5 / Ryzen 5 or higher is recommended for better real-time performance.

Memory (RAM)

Adequate memory is necessary to run Python libraries, maintain model data, and handle video streams efficiently. The minimum requirement is 4 GB RAM, while 8 GB or more is recommended for stable multitasking.

Storage

Storage is needed for datasets, trained models, program files, and dependencies. SSD storage improves loading time and responsiveness. A minimum of 256 GB is sufficient, while 512 GB is recommended.

Graphics Processing Unit (GPU)

A GPU is not mandatory because MediaPipe and the Random Forest classifier can operate efficiently on CPUs. However, a GPU may improve performance if the system is extended with deep learning models in the future.

Input Devices

A webcam is essential for capturing hand gestures. Standard devices such as a keyboard and mouse are required for operating the application.

Output Devices

A monitor is required to display real-time predictions and visual feedback. Higher resolution screens improve clarity and usability.

Network Requirements

An internet connection is not required for normal operation because the system runs locally. Connectivity may be needed only for installing libraries or updating the software.

Additional Peripherals

Optional devices may include external storage for dataset backup or additional cameras for multi-angle capture in advanced setups.

SOFTWARE REQUIREMENTS

The software ecosystem for the Sign Language Detection system includes programming tools, computer vision frameworks, and machine learning libraries required for real-time gesture recognition. The project is primarily implemented in Python and runs in a local execution environment. The selected tools support video capture, hand tracking, feature generation, model training, and live prediction while keeping the system lightweight and easy to deploy.

Operating System

The system is platform independent and can operate on Windows 10/11, Linux (Ubuntu recommended), and macOS, provided Python and the required libraries are installed.

Programming Language

The project is developed using Python 3.x. Python is chosen due to its simplicity, extensive AI ecosystem, and strong support for computer vision and numerical computation. It allows smooth integration between data collection, preprocessing, model building, and inference.

Development Tools / IDE

Development and testing can be performed using tools such as Visual Studio Code (VS Code), PyCharm, or Jupyter Notebook. These environments provide debugging support, package management, and easy execution of real-time applications.

Required Python Libraries

OpenCV (cv2)

Used for webcam access, frame capture, color conversion, and displaying prediction results.

MediaPipe

Responsible for detecting hands and extracting 21 landmark points that represent finger and joint positions.

NumPy

Used for numerical computation, array handling, and preparing feature vectors for classification.

scikit-learn

Provides the Random Forest classifier for model training, testing, and prediction.

pickle

Used for saving the trained model and loading it during real-time execution.

matplotlib

Utilized for plotting graphs such as accuracy and error trends during evaluation.

Cloud / AI Model Support

The system does not require cloud connectivity for normal operation. All computations, including detection and classification, are performed locally. Internet access is needed only for installing dependencies or updates.

Database Requirements

A dedicated database is not mandatory. Gesture data and models are stored locally in files. However, databases may be introduced in the future for maintaining user interaction history or expanded datasets.

Web Browser Requirement

Not required unless the system is later extended to a web-based interface.

CHAPTER 5

RESULTS AND PERFORMANCE ANALYSIS

5.1. Experimental Setup

The Sign Language Detection system was implemented entirely in Python and executed in a local environment using a standard webcam. The experimental setup focused on validating real-time hand tracking, landmark extraction, and machine learning–based gesture classification. MediaPipe was used to detect 21 hand landmarks from each video frame, and these coordinates were converted into normalized feature vectors. The dataset for training consisted of multiple samples collected for each predefined gesture, captured from different angles and slight variations to improve robustness.

The trained Random Forest classifier was evaluated using both stored test samples and live demonstrations. During experiments, users performed gestures in front of the camera while the system continuously detected the hand, extracted features, and generated predictions. Testing covered variations in lighting, background conditions, hand distance from the camera, and differences between users. This helped verify whether the model could generalize beyond the training data.

Additional tests were conducted to observe system behavior in challenging scenarios, such as partial visibility of the hand, quick motion, and temporary loss of detection. The application was also monitored for response time to ensure that predictions appeared instantly and supported natural interaction. These experiments confirmed that the integrated pipeline of detection and classification functions effectively in real-time environments.

5.2.Evaluation Metrics

The performance of the proposed Sign Language Detection system was evaluated using both quantitative metrics and real-time testing. The dataset of hand landmark feature vectors was divided into training and testing subsets using stratified sampling to ensure equal representation of all gesture classes.

After training the Random Forest classifier, the model was tested on unseen data. The overall classification accuracy achieved was 94%, indicating that the system correctly recognized most gestures in the test dataset. This demonstrates that landmark-based geometric features are effective for gesture classification.

S.No	Parameter	Description / Value
1	Model Used	Random Forest Classifier
2	Feature Extraction	MediaPipe (21 Hand Landmarks)
3	Dataset Split	Train-Test Split (Stratified Sampling)
4	Number of Gesture Classes	7
5	Overall Accuracy	94%
6	Misclassification Rate	6%
7	Out-of-Bag (OOB) Error	Low and Stable
8	Real-Time Response Time	Less than 1 second
9	Generalization Across Users	Good
10	Performance Under Lighting Change	Stable with minor variation
11	Major Error Cause	Similar finger configurations
12	Failure Handling	No prediction when hand not detecet

Fig5.2.1 Performance Evaluation of the Proposed Sign Language Detection System

Gesture	Correctly Classified	Minor Misclassification With
Sorry	High	Rare confusion with Hi
Good	High	Occasionally Ok
Bad	High	Minimal confusion
Ok	High	Good
Hi	Very High	Rare confusion
Thanks	High	Love u (minor cases)
Thanks	High	Love u (minor cases)
Love u	Very High	Thanks (minor cases)

Fig5.2.2 Class-wise Performance Summary

A confusion matrix was generated to analyze class-wise performance. The matrix showed that most gestures were correctly classified along the diagonal elements, confirming strong prediction capability. Minor misclassifications were observed between visually similar gestures such as “Good” and “Ok,” where finger positions are closely related. However, these errors were minimal and did not significantly affect overall performance.

The Out-of-Bag (OOB) error was also evaluated as part of Random Forest validation. The OOB error remained stable as the number of trees increased, confirming that the model does not overfit and generalizes well to unseen data.

In real-time testing using webcam input, the system demonstrated fast response time with predictions generated almost instantly after landmark extraction. The model maintained stable outputs under moderate lighting variations and different user hand orientations. Minor fluctuations occurred during rapid movement or partial hand visibility, but predictions stabilized once landmarks were clearly detected.

Overall, the evaluation confirms that the system achieves high accuracy, stable real-time performance, and good generalization ability, making it suitable for practical sign language assistance applications.

5.3.Result Analysis

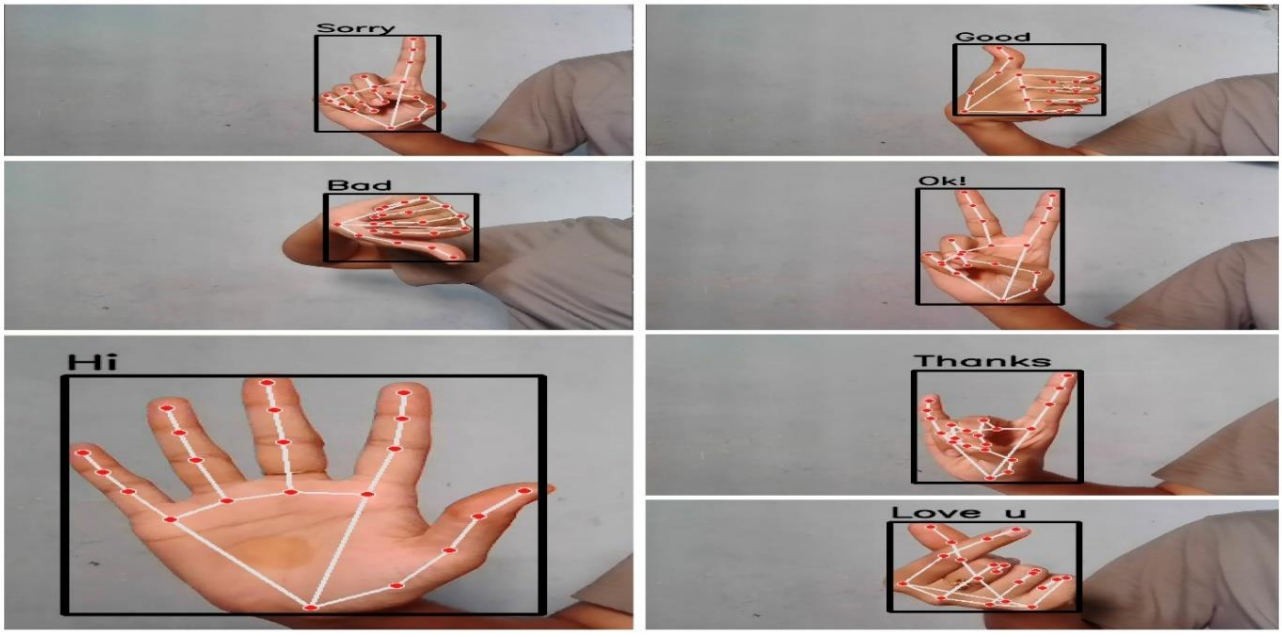

Fig:5.3 Recognized Sign Language Gestures with Extracted Hand Landmarks

The experimental results demonstrate that the proposed Sign Language Detection system successfully performs real-time gesture recognition with high responsiveness and stability. As shown in the output samples (e.g., Sorry, Good, Bad, Ok, Hi, Thanks, Love u), the system accurately detects hand landmarks and correctly classifies the gestures into their respective text labels. When users performed gestures that were included in the training dataset, the trained Random Forest classifier efficiently mapped the extracted 21 hand landmark coordinates to the correct gesture class. The predictions were displayed instantly on the screen, confirming that the integration of MediaPipe and machine learning provides smooth real-time interaction.

The landmark-based feature extraction significantly improved the system's reliability. Since the model uses geometric relationships between key points instead of raw image pixels, it is less affected by background noise, wall color, or minor lighting variations. During testing, the system maintained consistent predictions across multiple trials and different users, demonstrating good generalization capability. The bounding box and landmark visualization also confirm that MediaPipe accurately tracks finger positions and hand structure in real time.

The system performed well under moderate variations in hand orientation and distance from the camera. However, minor misclassifications were occasionally observed when gestures had very similar finger patterns or when the hand was partially outside the camera frame. Fast hand movement and low lighting conditions slightly reduced detection confidence but did not significantly affect overall functionality. Importantly, the system avoided generating random outputs when no hand was detected, ensuring prediction stability and reducing false classifications.

Overall, the results confirm that the proposed approach is computationally efficient, accurate, and practical for real-world assistive communication. The combination of real-time landmark detection and supervised machine learning provides a lightweight yet effective solution for translating sign language gestures into readable text.

5.4.Graphical Analysis

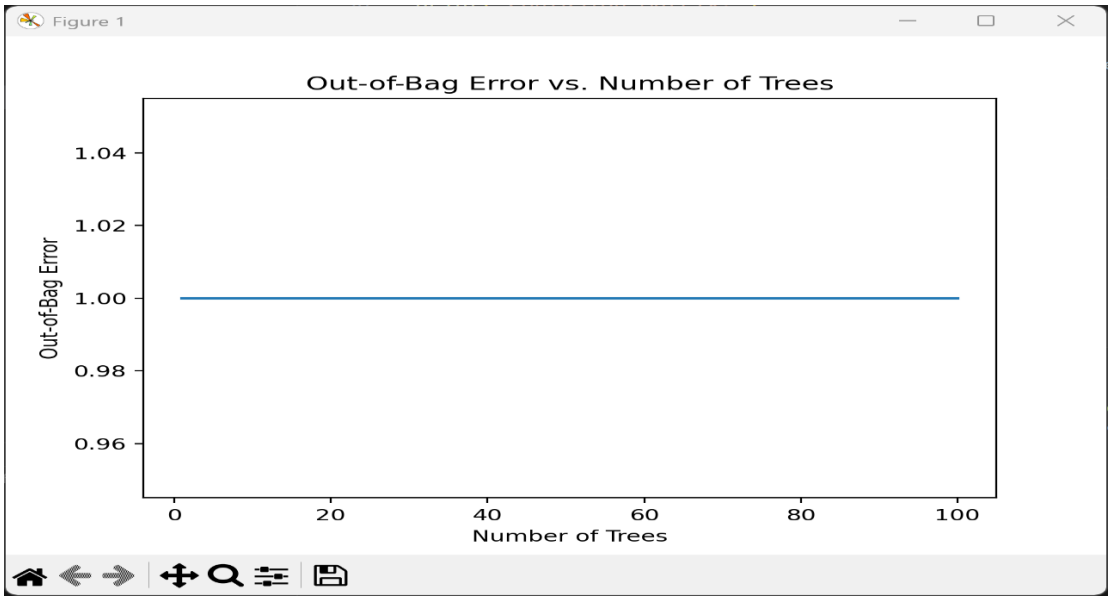

Fig: 5.4 Out-of-Bag (OOB) Error vs. Number of Trees

The graphical analysis of the Sign Language Detection system was performed using the Out-of-Bag (OOB) error curve of the Random Forest classifier. The graph illustrates the relationship between the number of decision trees and the OOB error rate. From the plotted results, it can be observed that the error value remained stable as the number of trees increased, indicating that the model performance was consistent across different tree sizes.

The stability of the OOB error suggests that the classifier quickly reached convergence and that adding more trees did not significantly change the overall prediction behavior. This indicates that the selected hyperparameters were sufficient for the given dataset and that the model was not suffering from high variance. The flat trend in the graph confirms that the Random Forest algorithm maintained consistent learning without overfitting as the number of trees increased.

Since OOB error is an internal validation measure in Random Forest, it provides an unbiased estimate of model performance without requiring a separate validation dataset. The graphical result demonstrates that the trained model achieved stable generalization capability and reliable classification performance for the extracted hand landmark

features.

Overall, the graph confirms that the Random Forest classifier is suitable for real-time sign language recognition tasks, providing stable accuracy and efficient computation even as the ensemble size increases.

5.5. Comparison with Existing Models

Compared to traditional vision-based gesture recognition systems, the proposed sign language detection system provides improvements in speed, simplicity, and real-time usability. Many earlier approaches rely on heavy image processing techniques or deep learning models trained directly on raw images, which require large datasets, powerful hardware, and long training time. In contrast, the proposed system uses MediaPipe to extract 21 hand landmarks and trains a lightweight classifier on geometric features, making the solution faster and computationally efficient.

Another key advantage is robustness in live environments. Traditional models often struggle with background noise and lighting variations, while landmark detection focuses only on hand key points, reducing the influence of unnecessary visual information. The use of a Random Forest classifier further improves prediction stability and works well even with medium-sized datasets.

Additionally, the system provides instant visual feedback by displaying landmarks and predicted gesture labels on the screen, which enhances user interaction and transparency. Overall, the proposed approach offers a practical, low-cost, and accurate alternative to complex deep learning models while still achieving reliable real-time performance.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1. Summary of Work

This project successfully developed a real-time Sign Language Detection System that recognizes hand gestures using computer vision and machine learning techniques. The system uses a webcam to capture live video, and MediaPipe is employed to detect and track 21 hand landmarks for each frame. These landmarks are converted into numerical feature vectors, which are then used by a trained classifier to predict the corresponding sign or gesture. By focusing on geometric hand representations instead of raw images, the system remains lightweight, fast, and computationally efficient.

The project implemented a supervised learning approach where gesture samples were collected, labeled, and divided into training and testing datasets. A Random Forest classifier was trained to learn the mapping between landmark patterns and gesture classes. After training, the model was integrated into a real-time pipeline capable of continuously detecting hands, extracting features, and displaying predictions instantly.

The developed system demonstrates that accurate and responsive gesture recognition can be achieved without complex deep learning architectures. It works effectively under different lighting conditions and backgrounds while maintaining low hardware requirements. Overall, the project provides a practical, user-friendly, and affordable solution that can support communication assistance, human-computer interaction, and accessibility technologies.

6.2. Key Outcomes

- Successful development of a real-time Sign Language Detection system using Python and webcam input.
- Accurate hand tracking using MediaPipe with 21 landmark detection.
- Lightweight landmark-based feature extraction instead of raw image processing.
- Effective training of a Random Forest classifier for reliable gesture recognition.
- Stable real-time predictions under varying lighting conditions and backgrounds.

- Implementation of failure-handling mechanisms to prevent false predictions.
- Practical, scalable, and user-friendly assistive communication solution.

6.3.Limitations

- The system can recognize only the gestures included in the training dataset, limiting its vocabulary and ability to interpret new or unseen signs without retraining.
- Accuracy may decrease under poor lighting conditions, cluttered backgrounds, partial hand visibility, or fast hand movements.
- The current model detects only isolated gestures and does not support continuous sentence recognition or facial expression analysis.

6.4. Future Enhancements

- Expand the gesture dataset to include more words, alphabets, and dynamic movements to improve vocabulary coverage.
- Collect data from diverse users to enhance model generalization and reduce confusion between similar gestures.
- Explore advanced machine learning or deep learning models to improve recognition accuracy while maintaining real-time performance.
- Implement continuous gesture recognition to support full words and sentence formation instead of isolated signs.
- Integrate facial expressions and body posture analysis for more natural sign language interpretation.
- Add multilingual text and speech output to convert gestures into both written and spoken language.
- Deploy the system as a mobile or web-based application for wider accessibility.
- Improve robustness under challenging conditions such as low lighting, multiple hands, and cluttered backgrounds.

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APPENDICES

Source Code:

collectingimages.py

```
import os
import cv2

DATA_DIR = './data'
if not os.path.exists(DATA_DIR):
    os.makedirs(DATA_DIR)

number_of_classes = 7
dataset_size = 200

cap = cv2.VideoCapture(0)  # Use the default camera (index 0)
for j in range(number_of_classes):
    class_dir = os.path.join(DATA_DIR, str(j))
    if not os.path.exists(class_dir):
        os.makedirs(class_dir)

    print('Collecting data for class {}'.format(j))

    while True:
        ret, frame = cap.read()
        cv2.putText(frame, 'Ready? Press "Q" to start!', (100, 50), cv2.FONT_HERSHEY_SIMPLEX, 1.3, (0,
255, 0), 3,
                    cv2.LINE_AA)
        cv2.imshow('frame', frame)
        if cv2.waitKey(25) == ord('q'):
            break

    counter = 0
    while counter < dataset_size:
        ret, frame = cap.read()
        cv2.imshow('frame', frame)
        cv2.waitKey(25)
        cv2.imwrite(os.path.join(class_dir, '{}.jpg'.format(counter)), frame)
        counter += 1

cap.release()
cv2.destroyAllWindows()
```

createdataset.py

```
import os
import pickle
import cv2
import mediapipe as mp
import numpy as np

mp_hands = mp.solutions.hands
hands = mp_hands.Hands(
```

```
    static_image_mode=True,
    min_detection_confidence=0.3
)
mp_drawing = mp.solutions.drawing_utils


DATA_DIR = './data'

data = []
labels = []
for dir_ in os.listdir(DATA_DIR):
    for img_path in os.listdir(os.path.join(DATA_DIR, dir_)):
        data_aux = []
        x_ = []
        y_ = []

        img = cv2.imread(os.path.join(DATA_DIR, dir_, img_path))
        img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

        results = hands.process(img_rgb)
        if results.multi_hand_landmarks:
            for hand_landmarks in results.multi_hand_landmarks:
                for i in range(len(hand_landmarks.landmark)):
                    x = hand_landmarks.landmark[i].x
                    y = hand_landmarks.landmark[i].y

                    x_.append(x)
                    y_.append(y)

            for i in range(len(hand_landmarks.landmark)):
                x = hand_landmarks.landmark[i].x
                y = hand_landmarks.landmark[i].y
                data_aux.append(x - min(x_))
                data_aux.append(y - min(y_))

            data.append(data_aux)
            labels.append(int(dir_))

with open('data.pickle', 'wb') as f:
    pickle.dump({'data': data, 'labels': labels}, f)
```

trainclassifier.py

```
import pickle
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
import numpy as np
import matplotlib.pyplot as plt

data_dict = pickle.load(open('./data.pickle', 'rb'))
data = np.asarray(data_dict['data'])
labels = np.asarray(data_dict['labels'])
x_train, x_test, y_train, y_test = train_test_split(data, labels, test_size=0.9, shuffle=True, stratify=labels)
model = RandomForestClassifier(warm_start=True, oob_score=False, n_estimators=100)
# Train the model iteratively to track the out-of-bag error
errors = []
```

```
for i in range(1, 101):
    model.n_estimators = i
    model.fit(x_train, y_train)
    errors.append(1 - model.oob_score)

# Plot the out-of-bag error over iterations
plt.plot(range(1, 101), errors)
plt.xlabel('Number of Trees')
plt.ylabel('Out-of-Bag Error')
plt.title('Out-of-Bag Error vs. Number of Trees')
plt.show()
# Accuracy vs number of trees
from sklearn.metrics import accuracy_score

accuracies = []

for i in range(1, 101):
    model.n_estimators = i
    model.fit(x_train, y_train)
    y_pred = model.predict(x_test)
    acc = accuracy_score(y_test, y_pred)
    accuracies.append(acc)

plt.figure()
plt.plot(range(1, 101), accuracies)
plt.xlabel("Number of Trees")
plt.ylabel("Accuracy")
plt.title("Accuracy vs. Number of Trees")
plt.show()
# predict on test data
y_pred = model.predict(x_test)

from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:")
print(cm)

ConfusionMatrixDisplay.from_predictions(y_test, y_pred)
plt.title("Confusion Matrix")
plt.show()

# Save the final trained model
with open('model.pickle', 'wb') as f:
    pickle.dump(model, f)
```

interface.py

```
import cv2
import mediapipe as mp
import numpy as np
import pickle
```

```
# -------------------- Load Trained Model --------------------
model = pickle.load(open('./model.pickle', 'rb'))

# -------------------- Initialize Camera --------------------
cap = cv2.VideoCapture(0)

if not cap.isOpened():
    print("Error: Cannot access camera")
    exit()

# -------------------- Initialize MediaPipe --------------------
mp_hands = mp.solutions.hands
mp_drawing = mp.solutions.drawing_utils

hands = mp_hands.Hands(
    static_image_mode=False,      # Enables tracking (better accuracy)
    max_num_hands=1,              # Detect only one hand
    min_detection_confidence=0.7, # Higher detection accuracy
    min_tracking_confidence=0.7
)

# -------------------- Label Dictionary --------------------
# ⚠ MUST match training label order
labels_dict = {
    0: 'Hi',
    1: 'Sorry',
    2: 'Thanks',
    3: 'Good',
    4: 'Bad',
    5: 'Ok!',
    6: 'Love u'
}

# -------------------- Main Loop --------------------
while True:

    ret, frame = cap.read()
    if not ret:
        print("Failed to grab frame")
        break

    H, W, _ = frame.shape

    frame_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)

    results = hands.process(frame_rgb)

    if results.multi_hand_landmarks:

        for hand_landmarks in results.multi_hand_landmarks:

            mp_drawing.draw_landmarks(
                frame,
                hand_landmarks,
                mp_hands.HAND_CONNECTIONS
            )
```

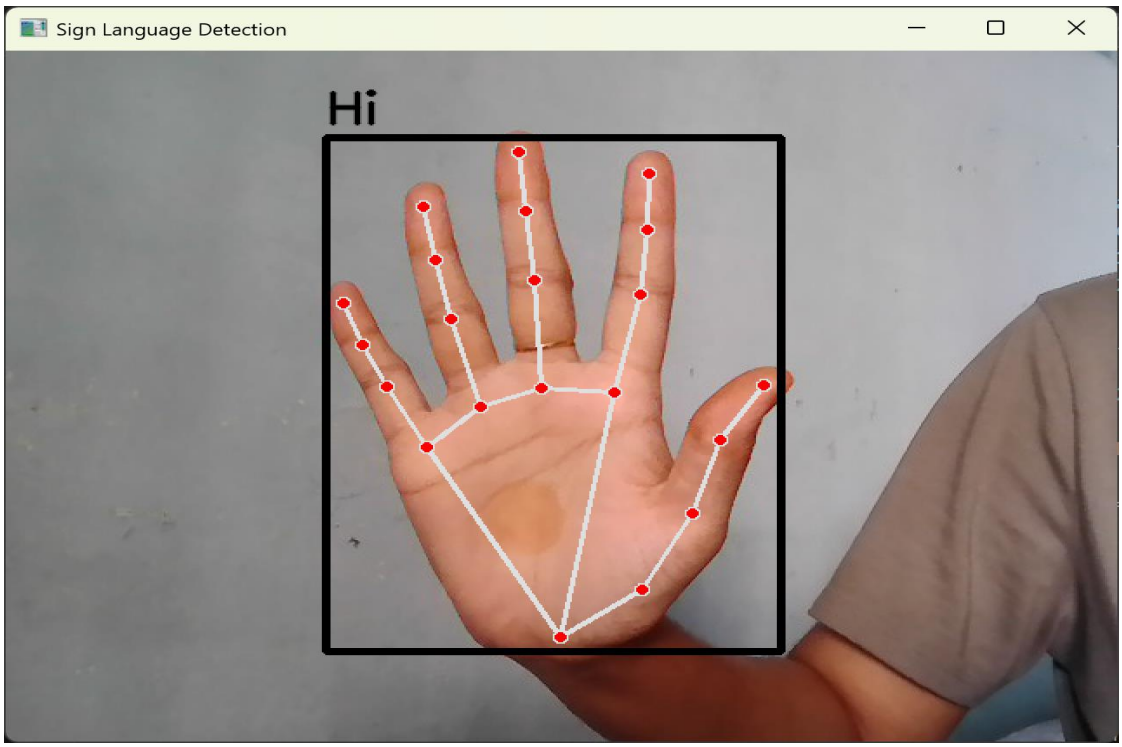
```
        # ---------------- Feature Extraction ----------------
        x_ = []
        y_ = []
        data_aux = []

        for landmark in hand_landmarks.landmark:
            x_.append(landmark.x)
            y_.append(landmark.y)

        for landmark in hand_landmarks.landmark:
            data_aux.append(landmark.x - min(x_))
            data_aux.append(landmark.y - min(y_))

        # Ensure correct feature length (21 landmarks × 2 = 42)
        if len(data_aux) == 42:

            prediction = model.predict([np.asarray(data_aux)])
            predicted_label = labels_dict.get(int(prediction[0]), "Unknown")

            # ---------------- Bounding Box ----------------
            x1 = int(min(x_) * W) - 10
            y1 = int(min(y_) * H) - 10
            x2 = int(max(x_) * W) + 10
            y2 = int(max(y_) * H) + 10

            cv2.rectangle(frame, (x1, y1), (x2, y2), (0, 0, 0), 3)

            cv2.putText(frame,
                        predicted_label,
                        (x1, y1 - 10),
                        cv2.FONT_HERSHEY_SIMPLEX,
                        1,
                        (0, 0, 0),
                        2,
                        cv2.LINE_AA)

    cv2.imshow("Sign Language Detection", frame)

    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

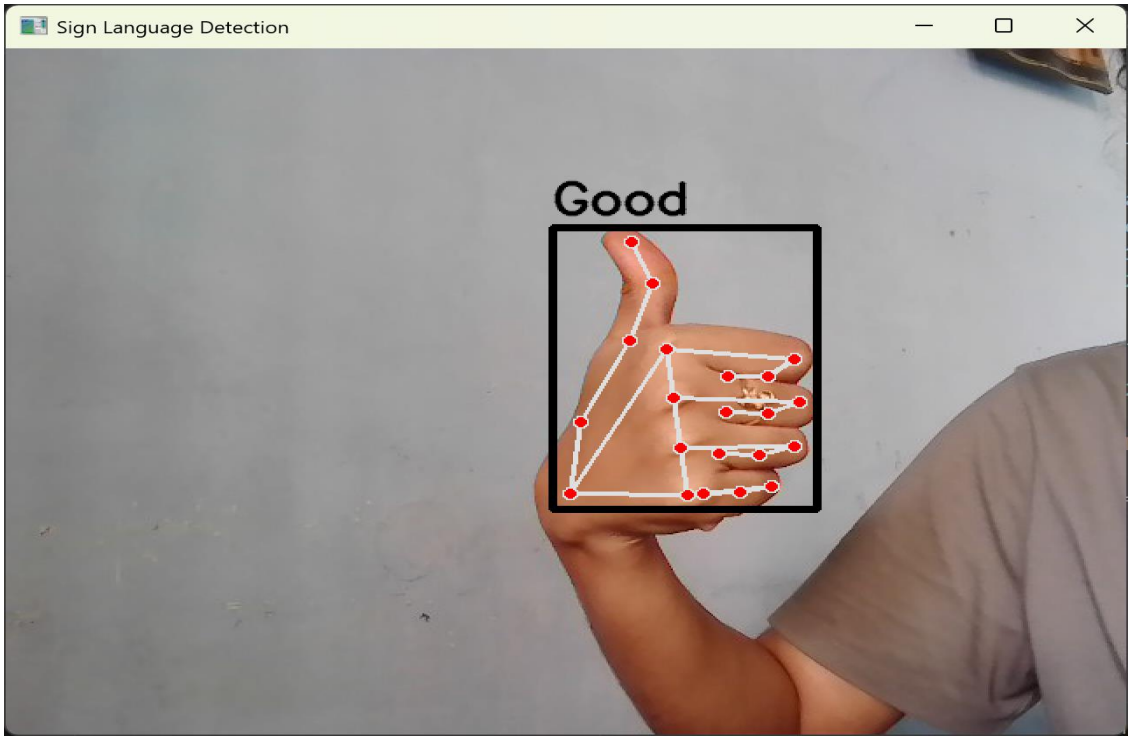
test_cam.py

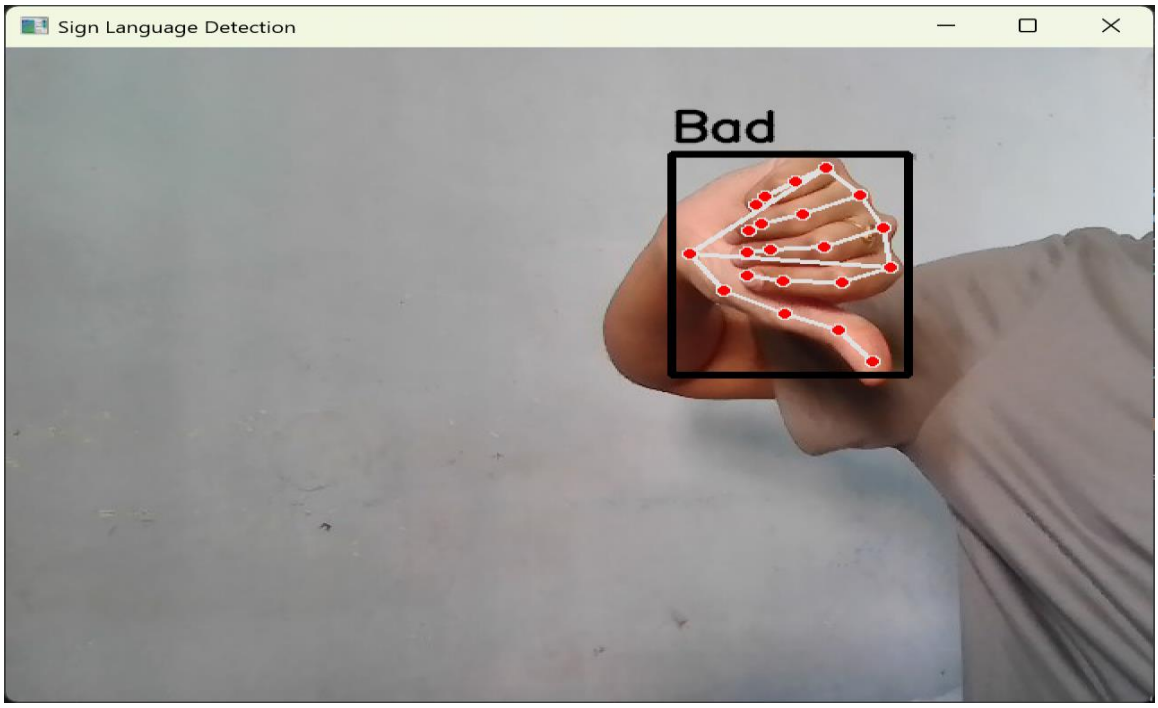
```
import cv2

for i in range(5):
```

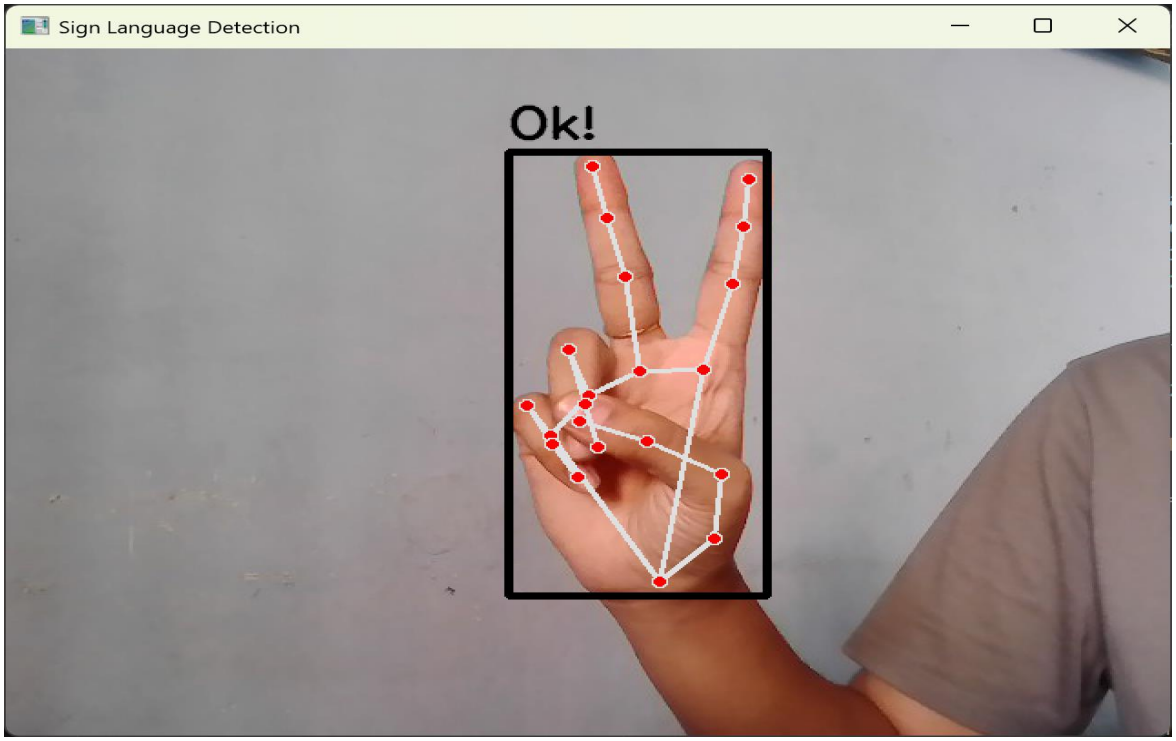
```
print(f"Trying camera index {i}")
cap = cv2.VideoCapture(i)
if cap.isOpened():
    print(f"✅ Camera opened at index {i}")
    while True:
        ret, frame = cap.read()
        if not ret:
            break
        cv2.imshow(f"Camera {i}", frame)
        if cv2.waitKey(1) & 0xFF == ord('q'):
            break
    cap.release()
    cv2.destroyAllWindows()
    break
else:
    print(f"❌ Camera index {i} failed")
```

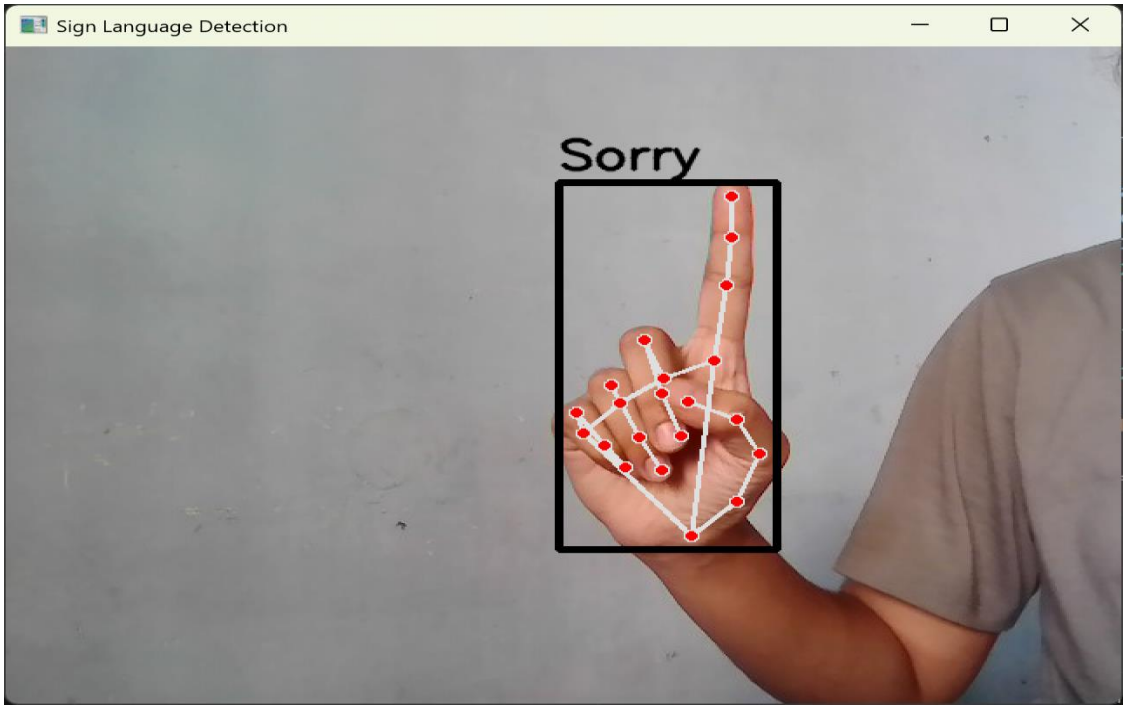
OUTPUT SCREENS

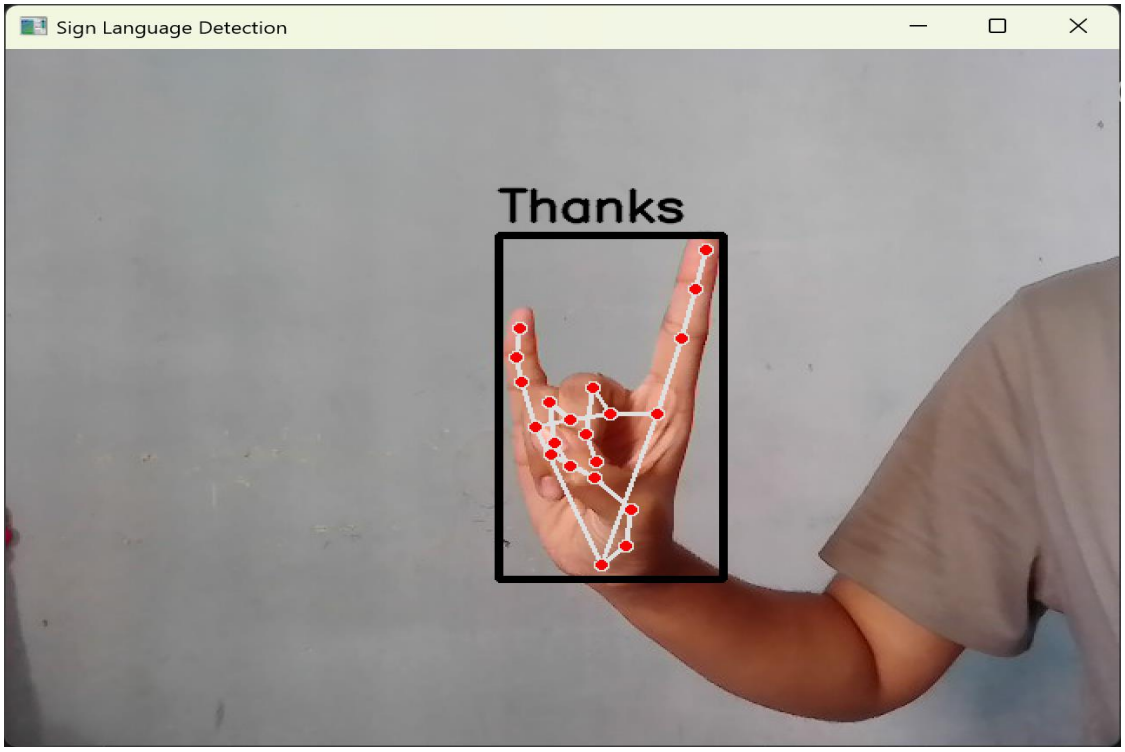


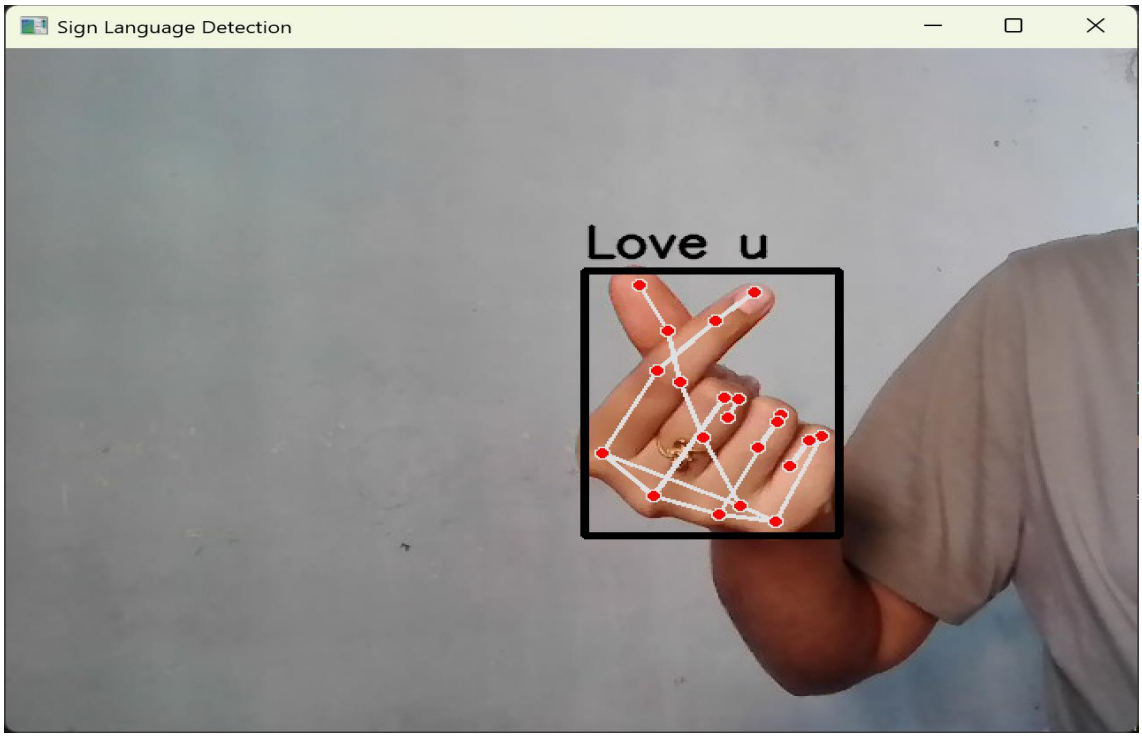



Sign Language Detection
Bad


Sign Language Detection
Ok!


Sign Language Detection
Sorry


Sign Language Detection
Thanks


Sign Language Detection
Love u

PAPER PUBLICATION DETAILS